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TURBULENCE MODELS 2024/2025 Final Assignment: Effect of Turbulence Models in CFD

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1 Introduction

Turbulence refers to chaotic, irregular, and unpredictable fluid motion characterized by rapid fluctuations in velocity, pressure, and other properties [10]. It occurs at high Reynolds numbers, where inertial forces dominate over viscous forces, leading to complex flow patterns [6]. Key features of turbulence include:

- **Irregularity**: Unsteady and seemingly random behavior.
- **Diffusivity**: Enhanced mixing and transport properties compared to laminar flow.
- **Dissipation**: Conversion of kinetic energy into heat due to viscosity.
- **Energy cascade**: Transfer of energy from large scales to small scales.

The cascade process describes how kinetic energy is transferred between scales in a turbulent flow:

- **Large Eddies and Turbulent Mixing.** Energy from large-scale motions, such as eddies, is transferred to progressively smaller scales through vortex interactions. As these large eddies move across the flow, they carry small-scale disturbances with them. When they approach the interface between the turbulent and nonturbulent regions, they distort the interface into a highly convoluted shape. In addition to migrating across the flow, these large eddies have a long lifetime, often persisting over distances up to 30 times the flow's width. As a result, the state of the turbulent flow at any given point depends on the history of upstream conditions, making it impossible to fully describe the flow in terms of the local strain-rate tensor as is done in laminar flow [10].
- **Diffusivity.** At very small scales, energy is dissipated as heat due to viscous effects. Turbulent diffusion significantly amplifies the transfer of mass, momentum, and energy. As a result, the apparent stresses in turbulent flows are typically several orders of magnitude greater than those in corresponding laminar flows [10].

Turbulence is primarily driven by large, energy-containing eddies, which are chiefly responsible for the enhanced diffusivity and stresses observed in turbulent flows. Due to the long persistence of these large eddies, the diffusivity and stresses depend on the flow's history and cannot always be expressed solely in terms of local flow properties. While the small eddies eventually dissipate turbulent energy through viscous effects, the rate of this dissipation is determined by how quickly energy is transferred from the largest eddies. These factors are crucial considerations in the development of any effective turbulence model [10].

Turbulence modeling provides a framework to approximate the effects of chaotic fluid motion without resolving all scales, crucial for practical computational fluid dynamics (CFD). Below is a

theoretical analysis of key turbulence models, emphasizing their foundational approaches and distinctions.

1.1 Kolmogorov law

It begins by noting that the cascade process present in all turbulent flows involves a transfer of turbulence kinetic energy (per unit mass), k, from larger eddies to smaller eddies. Dissipation of kinetic energy to heat through the action of molecular viscosity occurs at the scale of the smallest eddies. Because small scale motion tends to occur on a short time scale, we can reasonably assume that such motion is independent of the relatively slow dynamics of the large eddies and of the mean flow. Hence. the smaller eddies should be in a state where the rate of receiving energy from the larger eddies is very nearly equal to the rate at which the smallest eddies dissipate the energy to heat. This is one of the premises of Kolmogorov's (1941) universal equilibrium theory [10]. Hence, the motion at the smallest scales should depend only upon: (a) the rate at which the larger eddies supply energy, $\varepsilon = -dk/dt$, and (b) the kinematic viscosity, ν.
Kolmogorov scales of length (η), time (τ) and velocity (v) are

$$\eta \equiv (\nu^3/\varepsilon)^{1/4}, \qquad \tau \equiv (\nu/\varepsilon)^{1/2}, \qquad v \equiv (\nu\varepsilon)^{1/4}. \tag{1}$$

Since turbulence contains a continuous spectrum of scales, it is often convenient to do the analysis in terms of the spectral distribution of energy. In general, a spectral representation is a Fourier decomposition into wavenumbers, κ, or, equivalently, wavelengths, $\lambda = 2\pi/\kappa$. If $E(\kappa)d\kappa$ is the turbulence kinetic energy contained between wavenumbers κ and $\kappa + d\kappa$, one can say that

$$k = \int_0^\infty E(\kappa)d\kappa. \tag{2}$$

Recall that k is the kinetic energy per unit mass of the fluctuating turbulent velocity. Correspondingly, the energy spectral density or energy spectrum function, $E(\kappa)$, is related to the Fourier transform of k.

Observing that turbulence is so strongly driven by the large eddies, we expect $E(\kappa)$ to be a function of a length characteristic of the larger eddies, l, which is also known as the integral length scale [10].

In most turbulence-modeling analysis, it is assumed that there is a wide separation of scales, which means we implicitly assume l is very large compared to the Kolmogorov length scale, denoted as η.

$$l \gg \eta. \tag{3}$$

Utilizing notation mentioned above and following derivation from [10], the turbulence Reynolds number can be obtained:

$$Re_T \equiv \frac{k^{1/2}l}{\nu} \gg 1 \text{ because of equation 2.} \tag{4}$$

The existence of a wide separation of scales is a central assumption Kolmogorov made as part of his universal equilibrium theory. That is, he hypothesized that for very large Reynolds number, there is a range of eddy sizes between the largest and smallest for which the cascade process is independent of the statistics of the energy-containing eddies [10].

The idea is that a range of wavenumbers exists in which the energy transferred by inertial effects dominates, wherefore $E(\kappa)$ depends only upon ε and κ:

$$E(\kappa) = C_K \varepsilon^{2/3} \kappa^{-5/3}, \quad C_K \text{ is the Kologorov constant.} \tag{5}$$

Because inertial transfer of energy dominates, Kolmogorov identified this range of wavenumbers as the inertial subrange.

1.2 The Law of the Wall

The law of the wall is one of the most famous empirically-determined rela tionships in turbulent flows near solid boundaries. Measurements show that, for both internal and external flows, the streamwise velocity in the flow near the wall varies logarithmically with distance from the surface. This behavior is known as the Jaw of the wall.

Observation of high Reynolds number turbulent boundary layers reveals a useful, approximate description of the near-surface turbulence statistics. We find that effects of the fluid's inertia and the pressure gradient are small near the surface. Consequently, the statistics of the flow near the surface in a turbulent boundary layer are established by two primary mechanisms. The first is the rate at which momentum is transferred to the surface, per unit area per unit time, which is equal to the local shear stress, τ. The second is molecular diffusion of momentum, which plays an important role very close to the surface. Observations also indicate that the details of the eddies farther from the surface are of little importance to the near-wall flow statistics [10].

The validity of this approximate description improves with decreasing y/δ, where δ is the boundary-layer thickness. This is true because the ratio of typical eddy size far from the surface to eddy size close to the surface increases asy/δ decreases. In other words, since δ increases with Reynolds number, we find a wide separation of scales at high Reynolds numbers.

The quantity u_τ is known as the friction velocity, and is a velocity scale representative of veloci-

ties close to a solid boundary.

$$u_\tau \equiv \sqrt{\frac{\tau_w}{\rho}} \tag{6}$$

Where the surface shear stress, τ_w, is used in place of the local shear stress.

If one postulates that the mean velocity gradient, $\partial U/\partial y$, can be correlated as a function of u_τ, ν/u_τ and y, dimensional analysis, approximations and integration over y yield law of the wall:

$$u^+ = \frac{U}{u_\tau} = \frac{1}{\kappa}\ln y^+ + C, \quad y^+ = \frac{u_\tau y}{\nu}, \tag{7}$$

where C is a dimensionless integration constant. Correlation of measurements indicate $C \approx 5.0$ for smooth surfaces and $\kappa \approx 0.41$ for smooth and rough surfaces [10].

Three distinct regions can be observed on the plot u^+ over y^+: the viscous sublayer, the log layer and the defect layer. By definition, the log layer is the portion of the boundary layer where the sublayer and defect layer merge and the law of the wall accurately represents the velocity. It is not a distinct layer. Rather, it is an overlap region between the inner and outer parts of the boundary layer.

1.3 Zero-Equation Models

By the early 1950's, four main categories of turbulence models had evolved:

- Algebraic (Zero-Equation) Models
- One-Equation Models
- Two-Equation Models
- Stress-Transport Models

The simplest of all turbulence models are described as algebraic. These models use the Boussinesq eddy-viscosity approximation to compute the Reynolds stress tensor as the product of an eddy viscosity and the mean strain-rate tensor. For computational simplicity, the eddy viscosity, in turn, is often computed in terms of a mixing length that is analogous to the mean free path in a gas. In contrast to the molecular viscosity, which is an intrinsic property of the fluid, the eddy viscosity (and hence the mixing length) depends upon the flow. Because of this, the eddy viscosity and mixing length must be specified in advance, most simply, by an algebraic relation between eddy viscosity and length scales of the mean flow. Thus, algebraic models are, by definition, incomplete models of turbulence.

The Boussinesq eddy viscosity approximation is a foundational hypothesis in turbulence modeling, introduced by Joseph Boussinesq in 1877. It assumes that the additional momentum transfer

caused by turbulent eddies can be represented similarly to molecular viscosity, but with a so-called turbulent or eddy viscosity (ν_t or μ_t), which is not a fluid property but a property determined by the turbulence model.

Mathematically, the Boussinesq hypothesis states that the Reynolds stress tensor, which represents the effect of turbulence on the mean flow, is proportional to the mean rate of strain tensor:

$$-\overline{u_i' u_j'} = 2\nu_t S_{ij} - \frac{2}{3} k \delta_{ij}, \tag{8}$$

where

- $-\overline{u_i' u_j'}$ is the Reynolds stress tensor,
- ν_t is the turbulent (eddy) viscosity,
- S_{ij} is the mean strain rate tensor,
- k is the turbulent kinetic energy,
- δ_{ij} is the Kronecker delta.

While the Boussinesq approximation is widely used and often sufficiently accurate for engineering applications, it is a simplification. It assumes turbulence is isotropic and that the eddy viscosity concept is valid, which is not always true, especially in flows with strong anisotropy, rotation, or curvature.

The most widely known zero-equation model is the Prandtl mixing length model. The core assumption is that turbulent eddies transport momentum over a characteristic distance called the mixing length (l_m), analogous to the mean free path in kinetic theory. The turbulent (eddy) viscosity (ν_t) is then given by:

$$\nu_t = l_m^2 \left| \frac{dU}{dy} \right|, \tag{9}$$

where dU/dy is the mean velocity gradient perpendicular to the wall.

The mixing length is typically modeled as proportional to the distance from the wall in boundary layers:

$$l_{mix} = 0.41y. \tag{10}$$

This model provides a simple way to estimate turbulence effects but has several limitations.

Limitations

- The model does not account for the varying length scales in different flow regions, such as near walls or in free shear flows.

- It does not correctly predict turbulence behavior in flows with strong adverse pressure gradients (e.g., flow separation).
- The model assumes instantaneous response of turbulence to velocity gradients, but real turbulence has memory effects.
- It does not explicitly model the anisotropic nature of turbulence (doesn't take into account Reynolds stresses), making it inaccurate for complex 3D flows.

History of the Correlations for the Mixing Length

Close to solid boundary, the mixing length is approximately linear with distance from the surface, y. Measurements indicate that

$$l_{mix} \approx 0.41y. \tag{11}$$

This turns out to be a reasonably good approximation over a limited portion of a turbulent wall flow. For free shear flows such as jets, wakes and mixing layers, the mixing length is proportional to the width of the layer, δ. The mixing-length model fails to provide close agreement with measured skin friction for boundary layers. Since the mixing length was first postulated, considerable effort has been made aimed at finding a suitable prescription for boundary-layer computations.

The first key modification was devised by Van Driest (1956) who proposed that the mixing length should be multiplied by a damping function. This modification was made mainly as a good fit to data. The expression is following:

$$l_{mix} = 0.41y[1 - e^{-y^+/A_0^+}], \tag{12}$$

where $A_0^+ = 26$. Aside from the primary need to improve predictive accuracy, the Van Driest modification improves our description of the Reynolds stress in the limit $y \to 0$.

The second key modification was made by Clauser (1956) who addressed the proper form of the eddy viscosity in the defect layer. He specified

$$\nu_{T_0} = \alpha U_e \delta^*, \tag{13}$$

where ν_{T_0} is the kinematic eddy viscosity in the outer part of the layer, δ^* is the displacement thickness, U_e is the velocity at the edge of the layer, and α is the closure coefficient.

The third key modification is due to Corrsin and Kistler (1954) and Klebanoff (1954) as a corollary result of their experimental studies of intermittency. They found that approaching the free stream from within the boundary layer, the flow is not always turbulent. Rather, it is sometimes laminar and sometimes turbulent, i.e., it is intermittent. Their measurements indicate that for smooth

walls, the eddy viscosity should be multiplied by

$$F_{Kleb}(y;\delta) = [1 + 5.5(\frac{y}{\delta})^6]^{-1}, \tag{14}$$

where δ is the boundary-layer thickness.

The Cebeci-Smith model is a two-layer model with ν_T given by separate expressions in each layer. In terms of the normal distance from the nearest solid boundary, y, the eddy viscosity is

$$\nu_T = \begin{cases} \nu_{T_i}, & y \leq y_m \\ \nu_{T_0}, & y > y_m \end{cases} \tag{15}$$

where y_m is the smallest value of y for which $\nu_{T_i} = \nu_{T_0}$. Inner layer:

$$\nu_{T_i} = l_{mix}^2 \left[\left(\frac{\partial U}{\partial y} \right)^2 + \left(\frac{\partial V}{\partial x} \right)^2 \right]^{1/2} \tag{16}$$

$$l_{mix} = 0.4y[1 - e^{-y^+/A^+}] \tag{17}$$

$$A^+ = 26[1 + y\frac{dP/dx}{\rho u_\tau^2}]^{-1/2}. \tag{18}$$

Outer layer:

$$\nu_{T_o} = 0.0168 U_e \delta_v^* F_{Kleb}(y;\delta) \tag{19}$$

$$\delta^* = \int_0^\delta (1 - U/U_e) dy. \tag{20}$$

The Baldwin-Lomax model was formulated for use in computations where boundary-layer properties such as δ, δ_v^* and U_e are difficult to determine. This situation often arises in numerical simulation of separated flows, especially for flows with shock waves. Like the Cebeci-Smith model, this is a two-layer model. Mixing length is only showing in inner layer viscosity calculations:

$$\nu_{T_i} = l_{mix}^2 |\omega| \tag{21}$$

$$l_{mix} = 0.40y[1 - e^{-y^+/A^+}] \tag{22}$$

$$A_0^+ = 26. \tag{23}$$

1.4 One-Equation Spalart-Allmaras Model

The Spalart-Allmaras model [621] is a one-equation model that solves a modeled transport equation for the kinematic eddy (turbulent) viscosity. The Spalart-Allmaras model was designed specifically for aerospace applications involving wall-bounded flows and has been shown to give good results for boundary layers subjected to adverse pressure gradients. It is also gaining popularity in turbomachinery applications [1].

The transported variable in the Spalart-Allmaras model, $\widetilde{\nu}$, is identical to the turbulent kinematic viscosity except in the near-wall (viscosity-affected) region. The transport equation for the modified turbulent viscosity $\widetilde{\nu}$ is

$$\frac{\partial}{\partial t}(\rho\widetilde{\nu}) + \frac{\partial}{\partial x_i}(\rho\widetilde{\nu}u_i) = G_\nu + \frac{1}{\sigma_{\widetilde{\nu}}}\left[\frac{\partial}{\partial x_j}\left\{(\mu+\rho\widetilde{\nu})\frac{\partial\widetilde{\nu}}{\partial x_j}\right\} + C_{b2}\rho\left(\frac{\partial\widetilde{\nu}}{\partial x_j}\right)^2\right] - Y_\nu + S_{\widetilde{\nu}} \tag{24}$$

where G_ν is the production of turbulent viscosity, and Y_ν is the destruction of turbulent viscosity that occurs in the near-wall region due to wall blocking and viscous damping. $\sigma_{\widetilde{\nu}}$ and C_{b2} are the constants and ν is the molecular kinematic viscosity. $S_{\widetilde{\nu}}$ is a user-defined source term. Note that the turbulence kinetic energy, k, is not calculated in the Spalart-Allmaras model [1].

The turbulent viscosity μ_t is computed from

$$\mu_t = \rho\widetilde{\nu}f_{\nu 1} \tag{25}$$

$$f_{\nu 1} = \frac{\chi^3}{\chi^3 + C_{\nu 1}}, \qquad C_{\nu 1} = 0.71 \tag{26}$$

$$\chi \equiv \frac{\widetilde{\nu}}{\nu} \tag{27}$$

1.5 Standard $k - \varepsilon$

Two-equation turbulence models allow the determination of both, a turbulent length and time scale by solving two separate transport equations. It is a semi-empirical model, and the derivation of the model equations relies on phenomenological considerations and empiricism. The standard $k - \varepsilon$ model is a model based on model transport equations for the turbulence kinetic energy k and its dissipation rate ε. The model transport equation for is derived from the exact equation, while the model transport equation for was obtained using physical reasoning and bears little resemblance to its mathematically exact counterpart.

In the derivation of the $k - \varepsilon$ model, the assumption is that the flow is fully turbulent, and the effects of molecular viscosity are negligible. This model is therefore valid only for fully turbulent

flows. The turbulence kinetic energy, k, and its rate of dissipation, ε, are obtained from the following transport equations [3]:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + G_k + G_b - \rho\varepsilon - Y_M + S_k \tag{28}$$

$$\frac{\partial}{\partial t}(\rho \varepsilon) + \frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon}G_b) - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} + S_\varepsilon \tag{29}$$

In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are constants. σ_k and σ_ε are the turbulent Prandtl numbers for k and ε, respectively. and are user-defined source terms.

$C_{1\varepsilon}$=1.44, $C_{2\varepsilon}$=1.92, and C_μ=0.09, σ_k=1.0 and σ_ε=1.3 default values have been determined from experiments for fundamental turbulent flows including frequently encountered shear flows like boundary layers, mixing layers and jets as well as for decaying isotropic grid turbulence. They have been found to work fairly well for a wide range of wall-bounded and free shear flows.

The turbulent viscosity, μ_t, is computed by combining as follows:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \tag{30}$$

1.6 Standard $k - \omega$

The standard $k - \omega$ model in Ansys Fluent is based on a $k - \omega$ model proposed by Wilcox, which incorporates modifications for low-Reynolds number effects, compressibility, and shear flow spreading. One of the weak points of the 1998 Wilcox model is the sensitivity of the solutions to values for k and ω outside the shear layer (freestream sensitivity), which can have a significant effect on the solution, especially for free shear flows [4]. The turbulence kinetic energy, k, and the specific dissipation rate, ε, are obtained from the following transport equations:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j}\left(\Gamma_k \frac{\partial k}{\partial x_j}\right) + G_k - Y_k + S_k + G_b \tag{31}$$

$$\frac{\partial}{\partial t}(\rho\omega) + \frac{\partial}{\partial x_i}(\rho\omega u_i) = \frac{\partial}{\partial x_j}\left(\Gamma_\omega \frac{\partial \omega}{\partial x_j}\right) + G_\omega - Y_\omega + S_\omega + G_{\omega b} \tag{32}$$

In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_ω represents the generation of ω. Γ_k and Γ_ω represent effective diffusivity of k and ω, respectively. Y_k and Y_ω represent the dissipation of k and ω due to the turbulence. S_k and S_ω are user-defined source terms. G_b and $G_{\omega b}$ account for buoyancy.

The turbulent viscosity, μ_t, is computed by combining as follows:

$$\mu_t = \alpha^* \frac{\rho k}{\omega}, \qquad \alpha^* = 1.0. \tag{33}$$

1.7 Shear-Stress Transport (SST) $k - \omega$

The SST $k - \omega$ model accounts for the transport of the turbulence shear stress in the definition of the turbulent viscosity. These features make the SST model more accurate and reliable for a wider class of flows (for example, adverse pressure gradient flows, airfoils, transonic shock waves) than the above mentioned standard models [2]. Other models fail to properly predict the onset and amount of flow separation from smooth surfaces. The main reason is that other models do not account for the transport of the turbulent shear stress. This results in an overprediction of the eddy-viscosity. The proper transport behavior can be obtained by a limiter to the formulation of the eddy-viscosity:

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{max[\frac{1}{\alpha^*}, \frac{SF_2}{a_1 \omega}]}, \tag{34}$$

where S is the strain rate magnitude and α^* is defined as

$$\alpha^* = \alpha^*_\infty \left(\frac{\alpha^*_0 + Re_t/R_k}{1 + Re_t/R_k}\right), \quad Re_t = \frac{\rho k}{\mu\omega}, \quad R_k = 6, \quad \alpha^*_0 = \frac{0.072}{3}. \tag{35}$$

F_2 is given by

$$F_2 = \tanh \Phi_2^2, \qquad \Phi_2 = max\left[2\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega}\right], \tag{36}$$

where y is the distance to the next surface.

1.8 Explicit Algebraic Reynolds Stress Models (EARSM) $k-\omega$

Explicit Algebraic Reynolds Stress Models (EARSM) represent an extension of the standard two-equation models. They are derived from the Reynolds stress transport equations and give a non-linear relation between the Reynolds stresses and the mean strain-rate and vorticity tensors. Due to the higher order terms, many flow phenomena are included in the model without the need to solve transport equations for individual Reynolds stresses. The WJ-BSL-EARSM allows an extension of the baseline (BSL) turbulence model to capture the anisotropy of Reynolds stresses and secondary flows [5].

With EARSM, the Reynolds stresses are computed from the anisotropy tensor according to its definition:

$$\overline{u_i u_j} = k\left(a_{ij} + \frac{2}{3}\delta_{ij}\right), \tag{37}$$

where the anisotropy tensor a_{ij} is resolved as a solution of the following implicit algebraic matrix equation:

$$N\mathbf{a} = -A_1\mathbf{S} + (\mathbf{a}\Omega - \Omega\mathbf{a}) - A_2(\mathbf{aS} - \mathbf{Sa} - \frac{2}{3}tr\{\mathbf{aS}\}, \quad \text{with} \quad N = A_3 + A_4(P_k/\varepsilon). \tag{38}$$

The coefficients A_i in this matrix equation depend on the C_i-coefficients of the pressure-strain term in the underlying Reynolds stress transport model. Their values are selected here as A_1=1.245, A_2=0, A_3=1.8, A_4=2.25.

S_{ij} and Ω_{ij} denote the non-dimensional strain-rate and vorticity tensors, respectively. They are defined as:

$$S_{ij} = \frac{1}{2}\tau\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) \tag{39}$$

$$\Omega_{ij} = \frac{1}{2}\tau\left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i}\right) \tag{40}$$

where the time scale τ is given by

$$\tau = k/\varepsilon = 1/(C_\mu\omega), \qquad C_\mu = 0.09. \tag{41}$$

2 Project Objective

The primary objective of this project is to investigate and compare several widely used turbulence models in computational fluid dynamics (CFD), with an emphasis on their theoretical foundations and practical applications. The study focuses on zero-equation (mixing length), one-equation (Spalart-Allmaras), two-equation ($k-\varepsilon$, $k-\omega$, and $k-\omega$ SST), and Explicit Algebraic Reynolds Stress Models (EARSM). By analyzing the underlying assumptions, mathematical formulations, and treatment of near-wall turbulence-particularly in relation to the Law of the Wall-this project aims to highlight the major differences in the approaches adopted by each model. Ultimately, the goal is to provide a comprehensive understanding of the strengths and limitations of these models, thereby guiding the selection of appropriate turbulence modeling strategies for various engineering flow scenarios.

3 Simulation Setup

This chapter will present four different geometry models used in simulations, as well as the simulation setups for each of the casees.

3.1 Flat plate

The objective of this study is to model the turbulent boundary layer over a flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the zero equation, one-equation, two equation models and EARSM mdoel. In this exercise, the two-equation turbulence models, standard $k - \varepsilon$ and $k - \omega$, will be used. This simulation will validate the CFD results against theoretical predictions by comparing:

- Wall skin friction coefficient (C_f) at different momentum-thickness Reynolds numbers (4000 $< R_{e,\theta} <$ 13000).
- The law of the wall at $R_{e,\theta}$ = 10000 to verify the near-wall velocity distribution.
- In addition, one should also calculate results using two different grid resolutions and show the distributions of k, ω, ε and μ.

The aim is to evaluate the accuracy of the two-equation models and their ability to capture essential flow characteristics in a turbulent boundary layer.

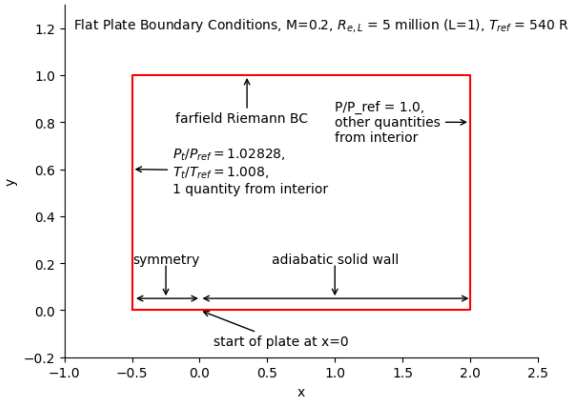
3.1.1 Flat plate simulation procedure

The study of boundary layer flow over a flat plate is fundamental in fluid mechanics, particularly in understanding transition from laminar to turbulent flow.

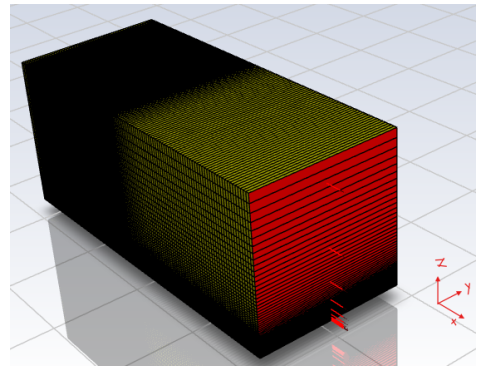
This problem is relevant in aerodynamics, heat transfer, and engineering applications where surface friction and turbulence effects play a crucial role in system performance. Computational Fluid Dynamics (CFD) provides a powerful tool to analyze such flows by solving the Navier-Stokes equations with turbulence models that approximate the behavior of small-scale eddies. Boundary conditions along with 2D representation of the plate is presented on the Figure 1a.

The following boundary conditions were applied:

- **Inlet:** A pressure inlet boundary is applied with pressure of 102.828.
- **Outlet:** A pressure outlet boundary condition is applied at the downstream edge of the computational domain, with the pressure set to zero gauge pressure. This allows for a smooth exit of the fluid, reducing artificial reflections.



(a) 2D flat surface with boundary conditions.



(b) Mesh representation in Fluent.

Figure 1: Flat plate simulation geometry and boundary conditions.

- **Flat plate:** The flat plate surface is modeled using a no-slip boundary condition, where both the streamwise and normal velocities are set to zero. This enforces the correct shear stress distribution necessary for boundary layer formation.
- **Far-field:** A far-field boundary condition is applied to the upper edge of the domain. This allows the flow to maintain free-stream conditions.

A CFD solver is used to simulate the flow with the following settings:

- Turbulence Model: Mixing-length (zero-equation), Spalart-Allmaras (one-equation), Standard $k-\omega$ and Standard $k-\varepsilon$ (two-equation), EARSM
- Grid: Structured mesh with refinement near the wall, coarser mesh 273$\times$193 and finer mesh 545$\times$385
- Solver: Pressure-based, steady-state
- Post-processing: A skin friction coefficient, the Law of the wall, and k, ω, μ, ε distributions

Definition of the quantities are given in the set of equations below.

$$Re_\theta = \frac{\rho_\infty U_\infty \theta}{\mu_\infty} \tag{42}$$

$$\theta = \int_0^\infty \frac{\rho}{\rho_\infty} \frac{u}{U_\infty} \left(1 - \frac{u}{U_\infty}\right) dy \tag{43}$$

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{44}$$

$$\tau_w = \mu_w \left(\frac{\partial u}{\partial y}\right)_w \tag{45}$$

$$u^+ = \frac{u}{v^*} = \frac{u}{\sqrt{\tau_w/\rho_w}} \tag{46}$$

$$y^+ = \frac{y v^*}{\nu_w} = \frac{y\sqrt{\tau_w/\rho_w}}{\nu_w} \tag{47}$$

3.2 Backward Step

The objective of this study is to model the backward step over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the two equation model. In this exercise, the one-equation Spalart-Allmaras model, two-equation turbulence models: standard $k - \varepsilon$, standard $k - \omega$ and SST $k - \omega$, and EARSM will be used. This simulation will validate the CFD results against theoretical predictions by comparing:

- Pressure coefficient C_p and wall skin friction coefficient (C_f) at the bottom wall of the plate.
- Velocity profiles and Reynolds stress distributions at streamwise positions $x/H = 1$ and $x/H = 6$.

This study not only helps assess the accuracy and limitations of different turbulence models in predicting separated turbulent flows, but also offers insight into model selection for engineering applications involving flow separation and reattachment.

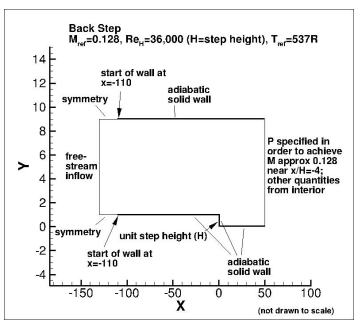
3.2.1 Backward Step Simulation Setup

Turbulent separated flows pose a significant challenge in computational fluid dynamics (CFD) due to their complex behavior involving flow detachment, recirculation zones, and reattachment. One

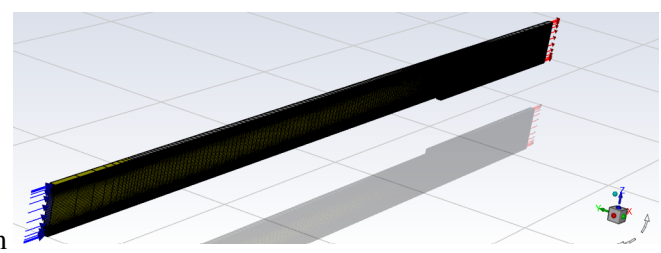
of the most widely studied benchmark cases for turbulent separation is the flow over a backward-facing step, which features sudden geometric expansion and flow separation downstream of the step.

This exercise aims to investigate the capability of three common Reynolds-averaged Navier-Stokes (RANS) turbulence models - Standard k-ε, Standard k-ω, and Shear Stress Transport (SST) - to, as well as one-equation Spalart-Allmaras, and EARSM Model to predict flow separation, reattachment, and associated turbulent phenomena in a backward-facing step configuration. The flow conditions, computational grid, and boundary conditions are provided by NASA Langley's turbulence modeling validation database.

The simulations are performed using these three turbulence models, and the results are analyzed in terms of their ability to reproduce key flow features. Boundary conditions along with 2D representation of the plate is presented on the Figure 2a and 3D geometry in ANSYS is presented on Fig. 2b.



(a) 2D surface with boundary conditions, taken from [9].



(b) Mesh representation in ANSYS Fluent.

Figure 2: The boundary conditions and geometry representation of the backward step.

Boundary conditions are presented below:

- **Inlet:** At the inlet, a uniform free-stream velocity U_∞ is prescribed to simulate the fully developed channel flow approaching the step. Turbulence quantities (turbulent kinetic energy k and either dissipation rate ε or specific dissipation rate ω) are also specified based on the selected turbulence model.
- **Outlet:** A pressure outlet boundary condition is applied at the downstream end of the domain, with the gauge pressure set to zero, allowing the fluid to exit the domain without artificial reflection of flow structures.
- **Bottom Wall:** The bottom wall, including the backward-facing step, is modeled using a no-slip boundary condition, enforcing zero velocity at the wall and enabling correct prediction of the boundary layer development, flow separation, and reattachment downstream of the step.

- **Top Wall:** The top wall is also treated as a no-slip wall (as in the experimental setup from NASA), to accurately capture the wall-bounded nature of the flow and match the confined channel flow condition. The computational domain is sufficiently extended upstream and downstream of the step to allow for fully developed inlet conditions and reattachment of the separated flow downstream. The mesh is refined in the vicinity of the step and near the walls to properly resolve the shear layer and boundary layer features.

A CFD solver is used to simulate the flow with the following settings:

- Turbulence Models: Spalart-Allmaras, Standard $k-\omega$, Standard $k-\varepsilon$, SST $k-\omega$, EARSM
- Grid: Structured mesh
- Post-processing: Pressure coefficient C_p, skin friction coefficient C_f, velocity and Reynold stress distributions at x/H = 1 and X/H = 6.

Definition of the quantities are given in the set of equations below.

$$\text{Pressure coefficient: } C_p = \frac{p - p_\infty}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{48}$$

$$\text{Friction coefficient: } C_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{49}$$

$$\text{Reynolds stresses: } -\rho\overline{u_i' u_j'} = \mu_T\left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} - \frac{2}{3}\frac{\partial \overline{u_k}}{\partial x_k}\right) - \frac{2}{3}\rho k \delta_{ij}, \quad \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{50}$$

$$\text{Incompressible flow: } \overline{u'u'} = -\frac{\mu_T}{\rho}\left(2\frac{\partial \overline{u}}{\partial x}\right) + \frac{2}{3}k \tag{51}$$

$$\overline{v'v'} = -\frac{\mu_T}{\rho}\left(2\frac{\partial \overline{v}}{\partial y}\right) + \frac{2}{3}k \tag{52}$$

$$\overline{u'v'} = -\frac{\mu_T}{\rho}\left(\frac{\partial \overline{u}}{\partial y} + \frac{\partial \overline{v}}{\partial x}\right) \tag{53}$$

$$\tag{54}$$

3.3 Jet Subsonic Plate

The objective of this study is to model a subsonic jet over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the one-equation Spalart Allmaras and two-

equation $k - \omega$ SST model with different inlet turbulence boundary conditions (Turbulent intensity 1%, 5% and 15%). This simulation will validate the CFD results against experimental results for:

- profiles of $u'v'/U_{jet}^2$, u/U_{jet}, v/U_{jet} and k/U_{jet}^2 at $x/D_{jet} = 2$, 5, 10, 15 and 20.

This study helps assess the influence of varying inlet turbulence intensity on the accuracy of jet flow predictions using the SST model by comparing key turbulence and velocity profiles with experimental data.

3.3.1 Subsonic Jet Simulation Setup

Subsonic jet flows are common in numerous engineering applications, including propulsion, HVAC systems, and industrial mixing processes. Accurate prediction of such flows is crucial, particularly in the near-field region where turbulence strongly influences the mixing characteristics. Computational Fluid Dynamics (CFD) provides a powerful tool for investigating jet behavior and validating turbulence models against experimental data.

This study investigates a subsonic round jet discharging from a nozzle over a flat plate, modeled using ANSYS Fluent. The axisymmetric jet configuration is representative of the ARN1 (Acoustic Research Nozzle 1, or ARN1) experimental setup. The primary objective is to assess the performance of the Shear Stress Transport (SST) k-ω and Spalart-Allamaras turbulence model under varying inlet turbulence intensities (1%, 5%, and 15%). The simulation results are compared with experimental data to evaluate their accuracy, particularly in terms of normalized velocity and turbulence profiles at selected streamwise locations. The EARSM model is not used in this simulation, since it cannot resolve the turbulence properties in 2D, which is the case with subsonic jet plate geometry.

The computational study was performed using ANSYS Fluent in a 2D axisymmetric configuration to model a subsonic jet issuing from a nozzle over a flat plate. The mesh domain represents a 1-degree slice rotated about the x-axis, with the jet centerline lying along the axis of symmetry ($y = 0$), presented on the Fig. 3. The simulation employed a steady-state, pressure-based solver. The velocity formulation was set to absolute, and the simulation was carried out under the assumption of axisymmetry to exploit geometric simplicity while capturing radial flow features.

Figure 3 shows the boundary conditions applied in the domain:

- **Inlet jet:** The jet inflow condition was applied with $P/P_{ref} = 1.19671$ and $T_t/T_{ref} = 1.0$ at $x = 0$, with meaning that initial supersonic and total gauge pressure were 119671 Pa at the inlet and temperature 295 K. These reference conditions are consistent with the ARN1 axisymmetric jet experiment. Turbulent intensity was set to 1%, 5%, and 15% for different cases.
- **Inlet freestream:** Velocity inlet with 3.43 m/s axial velocity was specified. The inlet was located upstream of the nozzle exit to allow full development of the jet.

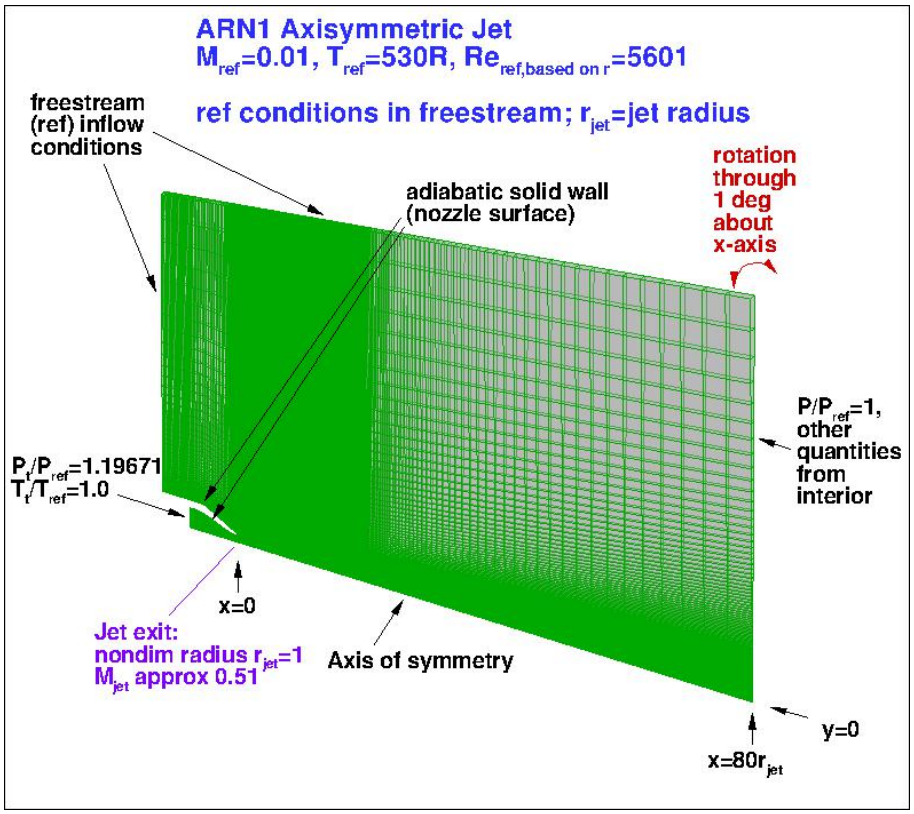


Figure 3: Computational domain and boundary conditions for the axisymmetric jet simulation, taken from [8].

- **Wall:** The nozzle surface was treated as an adiabatic solid wall, enforcing a no-slip condition and zero heat flux.
- **Symmetry:** The domain's lower boundary, along $y = 0$, was modeled as the axis of symmetry.
- **Outlet:** At the outlet ($x = 80r_{jet}$), a pressure outlet condition was applied with $P/P_{ref} = 1$. Set as a pressure outlet with the option *prevent reverse flow* enabled to avoid non-physical backflow and enhance solution stability.

The SST (Shear Stress Transport) k-ω and one-equation Spalart-Allmaras model were selected for turbulence modeling. Three different inlet turbulence intensities-1%, 5%, and 15%-were tested to examine their influence on the jet's development and agreement with experimental data.

Postprocessing was performed in Ansys Fluent and Python to extract relevant flow quantities along radial profiles at several downstream locations from the jet exit. The primary objective was to evaluate the influence of inlet turbulence intensity on the accuracy of predicted velocity and turbulence characteristics by comparing simulation results with experimental data.

The following non-dimensional quantities were extracted and plotted at $x/D_{jet} = 2$, 5, 10, 15, and 20:

- Normalized axial velocity: $\frac{u}{U_{jet}}$
- Normalized radial velocity: $\frac{v}{U_{jet}}$
- Normalized Reynolds shear stress: $\frac{u'v'}{U_{jet}^2}$
- Normalized turbulent kinetic energy: $\frac{k}{U_{jet}^2}$

The jet exit velocity U_{jet} was determined from the simulation by taking the value of the velocity magnitude along the centerline (axis of symmetry) at the nozzle exit ($x = 0$). The maximum value of the velocity magnitude at this location was taken as U_{jet}. The quantities were calculated using custom-defined field functions in Fluent as follows:

$$\frac{u}{U_{jet}} = \frac{\text{axial-velocity}}{U_{jet}} \tag{55}$$

$$\frac{v}{U_{jet}} = \frac{\text{radial-velocity}}{U_{jet}} \tag{56}$$

$$\frac{k}{U_{jet}^2} = \frac{\text{turb-kinetic-energy}}{U_{jet}^2} \tag{57}$$

$$\frac{u'v'}{U_{jet}^2} = \frac{-\mu_t}{\rho}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\frac{1}{U_{jet}^2} \tag{58}$$

The results from the CFD simulations were plotted alongside experimental data provided in a structured dataset, allowing direct comparison at each downstream location.

3.4 Trubulent Mixing Layer

In this part, the simulation of a turbulent mixing layer is performed using five different turbulence models: the Spalart-Allamaras, the $k - \varepsilon$, $k - \omega$ standard, $k - \omega$ SST models. The goal is to assess the performance of these turbulence models by comparing their predictions against experimental measurement data and theoretical expectations. The experimental data, computational grid, and boundary conditions necessary for the simulation are provided at:
`https://turbmodels.larc.nasa.gov/delvilleshear_val.html`.

The specific objectives of the homework are:

- To simulate the turbulent mixing layer using different turbulence models.

- To compare the computed CFD results with theoretical and experimental results for:
  - Mean velocity profiles at streamwise locations x = 1 mm, x = 200 mm, and x = 650 mm.
  - Reynolds stress profiles at streamwise locations x = 200 mm and x = 650 mm.

3.4.1 Turbulent Layer Simulation Setup

The simulation of turbulent mixing layers plays a crucial role in understanding the fundamental behavior of shear-driven flows, which are common in many engineering applications such as jet propulsion, combustion systems, and atmospheric boundary layers. A mixing layer forms when two parallel streams of fluid with different velocities interact, leading to the development of a shear region characterized by strong velocity gradients and turbulent fluctuations.

The objective of this study is to evaluate and compare the ability of various turbulence models to accurately capture the key features of a turbulent mixing layer, particularly the growth of the shear layer and the distribution of turbulent stresses.

The computational study was performed using ANSYS Fluent in a 2D mixing layer validation case. The computational domain represents a two-dimensional planar mixing layer, where two streams with slightly different total pressures and temperatures are introduced into the domain through pressure boundary conditions, presented on the Fig. 4. The simulation employed a steady-state, pressure-based solver.

Figure 4 shows the boundary conditions applied in the domain:

- **Inlet:** The top inflow condition was applied with $P/P_{ref} = 1.01031$ and $T_t/T_{ref} = 1.00293$ at $x = -500$ mm, with meaning that initial supersonic and total gauge pressure were 101031 Pa at the top inlet and temperature 293.86 K. The bottom inflow condition was applied with $P/P_{ref} = 1.00299$ and $T_t/T_{ref} = 1.00085$ at $x = -300$ mm, with meaning that initial supersonic and total gauge pressure were 100299 Pa at the bottom inlet and temperature 293.25 K. These reference conditions are consistent with the 2D Mixing Layer Experiment.
- **Top and Bottom Boundaries:** Slip wall conditions are applied at the top and bottom boundaries, allowing tangential flow while preventing normal flow. No shear stress is imposed (adiabatic slip walls).
- **Walls:** The solid walls adjacent (red lines in the Fig. 5)to the shear layer are defined as adiabatic solid walls with no normal velocity (impermeable) but allowing tangential flow (slip condition).
- **Symmetry:** The domain's lower boundary, along $y = 0$, was modeled as the axis of symmetry.
- **Outlet:** At the outlet, a pressure outlet condition was applied with $P/P_{ref} = 1.0$.

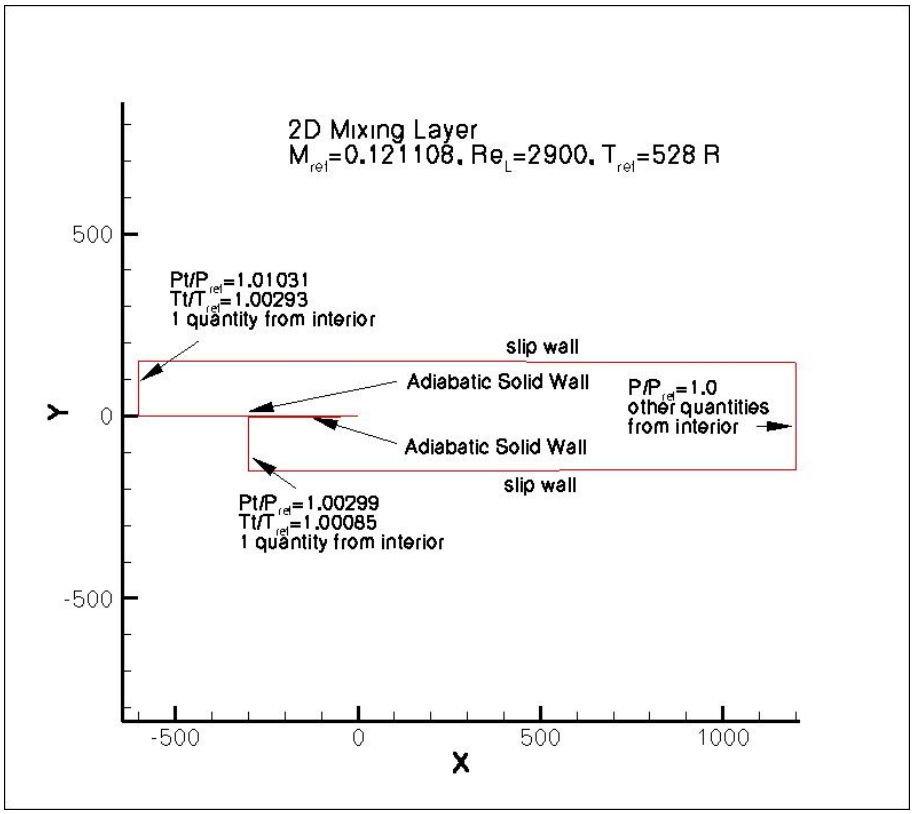


Figure 4: Computational domain and boundary conditions for the mixing boundary layer simulation case, taken from [7].

Second-order discretizations schemes are used for spatial derivatives to minimize numerical diffusion. Turbulence quantities at the inlet are initialized based on estimated turbulence intensity and length scales corresponding to the mixing layer conditions.

Postprocessing was performed in Ansys Fluent and Python to extract relevant flow quantities along z-axis profiles at several x-axis locations from the inlet. The primary objective was to evaluate the ability of various turbulence models to accurately capture the key features of a turbulent mixing layer.

The following non-dimensional quantities were extracted and plotted at $x = 1$, 200 and 650 mm:

- Streamwise velocity: u
- Reynolds stresses: $\overline{u'u'}, \overline{u'v'}, \overline{v'v'}$.

The Reynolds shear stresses were calculated using custom-defined field functions in Fluent as follows:

$$\overline{u'u'} = \frac{-\mu_t}{\rho}\left(2\frac{\partial u}{\partial x}\right) + \frac{2}{3}k \tag{59}$$

$$\overline{u'v'} = \frac{-\mu_t}{\rho}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \tag{60}$$

$$\overline{v'v'} = \frac{-\mu_t}{\rho}\left(2\frac{\partial v}{\partial y}\right) + \frac{2}{3}k \tag{61}$$

$$\tag{62}$$

The results from the CFD simulations were plotted alongside experimental data provided in a structured dataset, allowing direct comparison at each downstream location.

3.5 Rectangular Duct

The objective of this section is to simulate steady-state airflow through a rectangular duct at an inlet Reynolds number of 1205, using different turbulence models. The turbulence models to be applied are: Zero Equation model, Spalart-Allmaras model, Shear Stress Transport (SST) model, and Baseline Explicit Algebraic Reynolds Stress Model (BSL EARSM).

The simulation results will be compared to Direct Numerical Simulation (DNS) data, focusing on the tangential velocity vectors at the duct outlet. Additionally, the convergence time of each turbulence model will be analyzed and compared, providing insight into the computational efficiency and stability of each approach.

3.5.1 Rectangular Duct Simulation Procedure

The accurate prediction of turbulent flow behavior in ducts is essential for many engineering applications, such as ventilation systems, heat exchangers, and aerodynamic devices. Due to the complexity of turbulence, direct numerical simulation (DNS) is often used as a benchmark, but it is computationally expensive and impractical for most real-world problems. Therefore, various turbulence models have been developed to approximate the effects of turbulence with reasonable accuracy and computational cost.

This study aims to simulate steady airflow inside a rectangular duct using four different turbulence modeling approaches: the Zero Equation model, the Spalart-Allmaras model, the Shear Stress Transport (SST) model, and the Baseline Explicit Algebraic Reynolds Stress Model (BSL EARSM). The flow conditions are set at an inlet Reynolds number of 1205, with air at 25°C and 101.130 kPa, and a duct cross-sectional area of 4 cm^2.

The results from these turbulence models will be compared against DNS data, with a particular focus on the tangential velocity vectors at the duct outlet. Furthermore, the convergence time for each turbulence model will be evaluated and discussed to assess the trade-offs between computational efficiency and simulation accuracy.

The computational domain consists of a rectangular duct with a cross-sectional area of 4 cm^2. A structured mesh was generated to discretize the domain, ensuring sufficient refinement near the walls to capture boundary layer effects, presented on the Fig. 5. The simulations were performed under steady-state conditions. Appropriate solver settings were used for each turbulence model to ensure numerical stability, and convergence was monitored based on residual values, requiring at least a drop of 1×10^{-4} for all residuals.

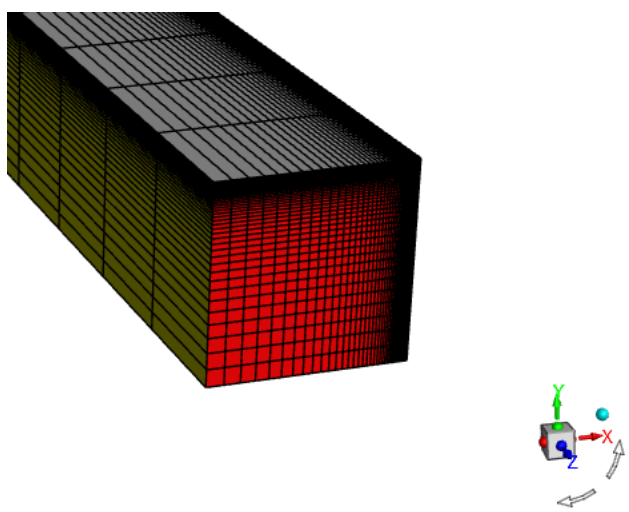


Figure 5: Computational domain and mesh for the rectangular duct in ANSYS Fluent.

The following boundary conditions were applied:

- **Inlet:** A uniform velocity profile was imposed, corresponding to a Reynolds number of 1205 based on the hydraulic diameter, which gives 0.94 m/s inlet velocity. Turbulence intensity and viscosity ratio were set according to standard inlet turbulence conditions, depending on the turbulence model. The fluid was modeled as incompressible air at a temperature of 25Â°C and an absolute pressure of 101.130 kPa.
- **Walls:** No-slip boundary conditions were imposed at all duct walls.
- **Symmetry:** The domain's lower boundary, along $y = 0$, was modeled as the axis of symmetry.
- **Outlet:** At the outlet, A fixed static pressure boundary condition was applied with gauge = absolute pressure of p = 101.130 kPa.

Four turbulence models were employed in the simulations: Zero Equation model (Turbulent Mixing-length), Spalart-Allmaras model, Shear Stress Transport (SST) model and Baseline Explicit Algebraic Reynolds Stress Model (BSL EARSM). The simulations were initialized using a standard procedure, where the initial velocity and turbulence fields were specified according to the inlet conditions.

After convergence, post-processing was performed by capturing vector plots of the velocity magnitude at the outlet. Run time data for each simulation was also collected to evaluate the computational cost associated with each turbulence model.

4 Results & Analysis

In this section, the results for each simulation case presented in previous chapter are discussed in this chapter. Results are presented respectively.

4.1 Flat Plate

Distributions of the wall skin friction coefficient C_f over different Reynolds numbers against theoretical values for this case are presented on Fig. 6. The Law of the Wall for different turbulence models against theoretical values is presented on the Fig. 7.

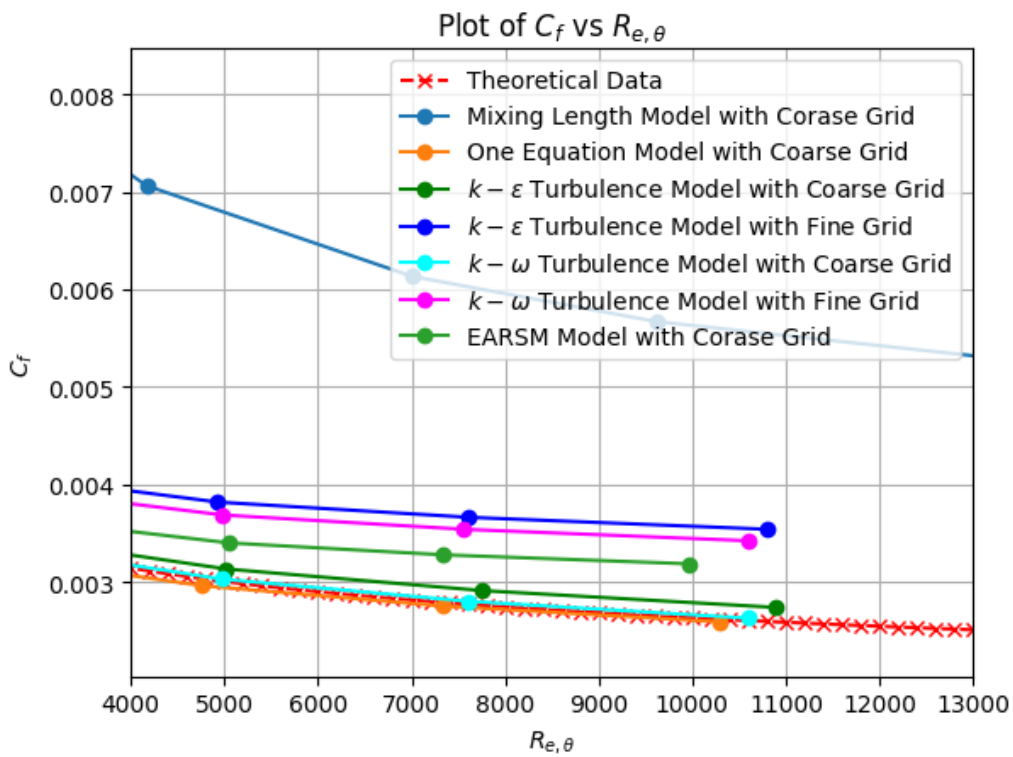


Figure 6: The wall skin friction coefficient over the Reylonds number for the range $4000 < R_e < 13000$ for different turbulence models using two different meshes.

It can be seen on Fig. 6 that mixing length model in this case overestimates/over-predicts the wall friction coefficient. One of the reasons is that mixing length model does not account for the viscous sublayer in turbulent boundary layers. It assumes that the turbulent eddy viscosity is always proportional to the velocity gradient, even very close to the wall. in reality, near-wall turbulence is damped by molecular viscosity, leading to a low turbulence production region. By ignoring this damping effect, the model predicts too much turbulence near the wall, leading to higher shear stress and

an overestimated C_f. One-Equation model results for C_f is showing better agreement with theoretical results across a range of Reynolds numbers than mixing length model, Fig. 6. It captures the development of the turbulent boundary layer more accurately, especially at higher Reynolds number values. One-model equation can predict pressure gradients and flow separation better due to its ability to account for turbulent transport.

From Fig. 6 can be seen that Standard $k-\omega$ and $k-\varepsilon$ two-equation turbulence models on coarser grid (273×139) reproduce more faithful values of C_f rather than on the finer grid (545×385) comparing to the theoretical values. Between $k-\omega$ and $k-\varepsilon$ on the coarse grid, $k-\omega$ gives the closest fit to the theoretical predictions. This can be explained by the models grid sensitivity and near-wall treatment: the $k-\omega$ mode is known for its better near-wall resolution becase ω governs the turbulent length scale close to the wall. On the other hand the $k-\varepsilon$ model is more accurate in the fully turbulent region but requires wall functions for near-wall treatment, which can introduce errors on finer grids if the first grid point is not placed correctly in the boundary layer. One other point is, in the coarse grid, the gradients may not be perfectly captured, but wall functions compensate for this, making the results closer to the theoretical predictions. In the fine grid, if the first cell height is not sufficiently small, the turbulence model might not handle the boundary layer correctly, leading to deviations.

In the same spirit, the law of the wall is calculated and plotted against theoretical values for $R_{e,\theta}$ = 10000. Result is depicted on Fig. 7. In this case, that Reynolds number shows around surface line x=0.6. It can be seen that only in viscous sublayer, the mixing length model and theoretical predictions are in good accord. Since turbulence is weak in the viscous sublayer, assuming a constant or very small mixing length does not introduce significant errors. The model fails in the viscous sublayer, where eddy viscosity should be much smaller. Since the model does not reduce eddy viscosity properly at low y, it predicts excessive momentum transfer, increasing wall shear stress and friction. One-Equation Model more accurately reproduces the log-law region and viscous sublayer. It captures the full structure of the boundary layer, including the buffer and outer regions.

The law of the wall is the best reproduced with $k-\omega$ model on the coarser grid (line in cyan). $k-\varepsilon$ model on both grids give very bad results in the viscous sublayer, but $k-\varepsilon$ model on coarse grid captures logarithmic layer behavior very well. It is because the $k-\varepsilon$ model is originally designed for high-Reynolds-number turbulence and does not perform well for small distances from the wall. However, on the fine grid, the incorrect near-wall resolution affects the entire profile, leading to deviations in both the viscous and logarithmic layers for both models. A finer grid should improve accuracy, but only if the turbulence model is correctly formulated for low y^+. If the grid is too fine for wall functions, but not fine enough for direct resolution of the viscous layer ($y^+ \approx$1), the model fails to provide correct near-wall predictions.

As part of the analysis of this case, the distributions of eddy viscosity for both models on both grids are calculated and presented on Fig. 8. The eddy viscosity ν_t represents the additional viscosity

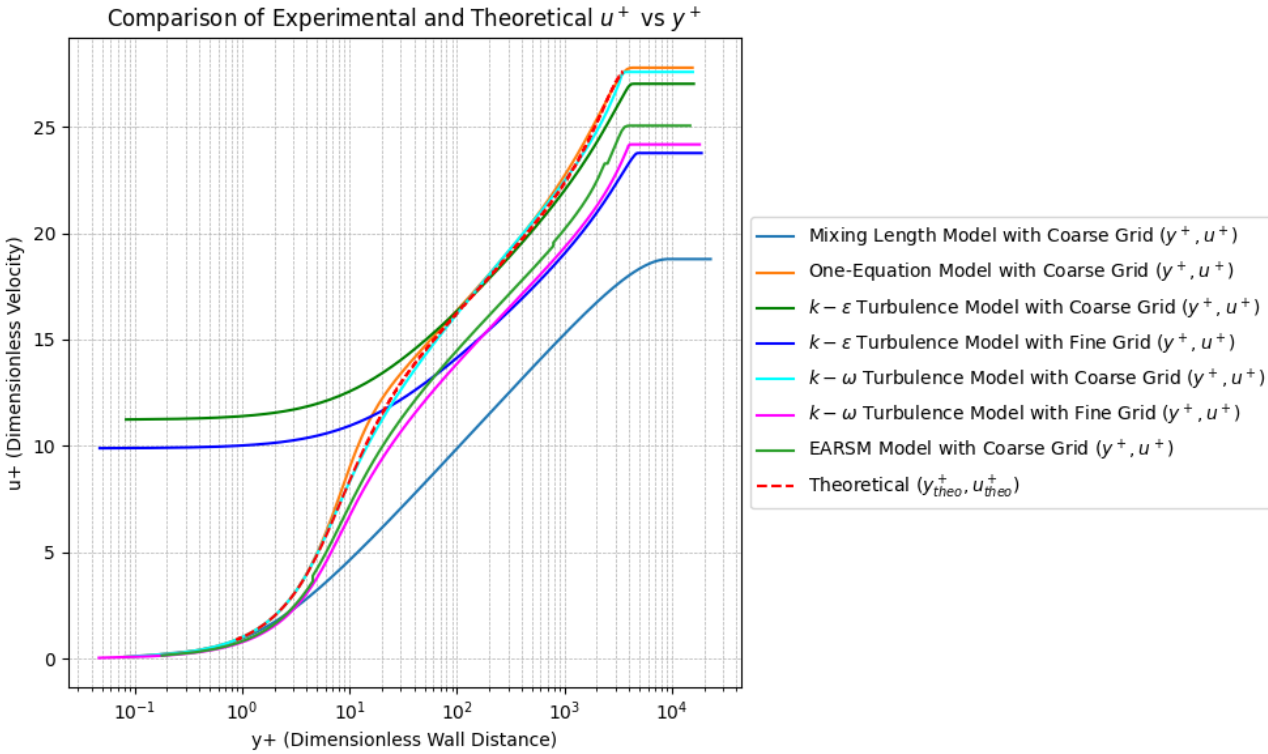
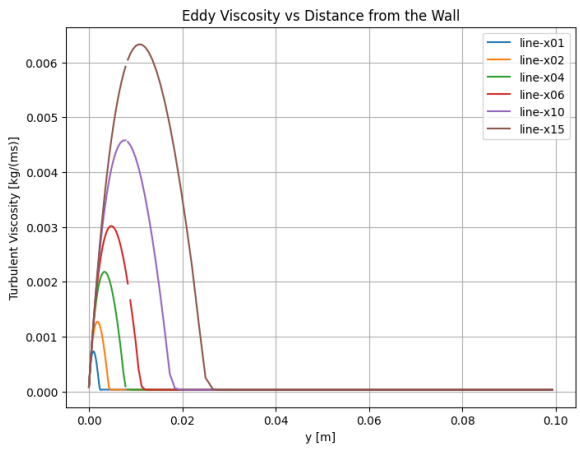


Figure 7: The law of the wall distributions for different turbulence models applied on two different meshes (R_e=10000).

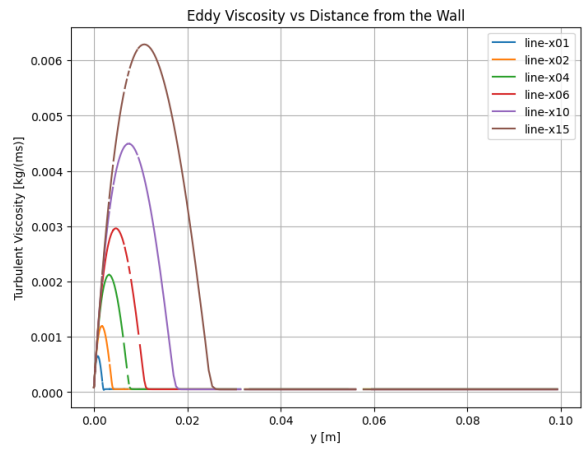
caused by turbulence, which enhances the momentum transfer. In the viscous sublayer, the flow is laminar-like, dominated by molecular viscosity, and turbulence is very weak. Since turbulence production is minimal in this region, $\nu_t \approx$0 near the wall.

In the buffer layer and logarithmic layer, turbulence production increases due to shear effects. The peak in turbulence kinetic energy and turbulence dissipation occurs in this region. The maximum eddy viscosity is found at the location where turbulence production is highest. As you move further away from the wall into the free-stream region, velocity gradients become small. Turbulence generation diminishes, and eddy viscosity gradually decreases to zero. In the fully developed outer region, turbulence is mainly convected away, and molecular viscosity dominates again.

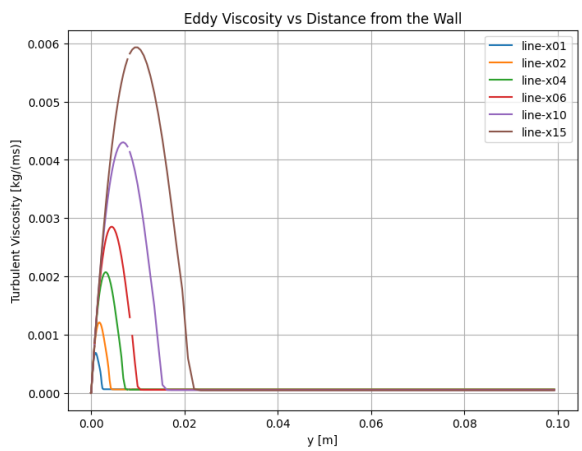
The maximum eddy viscosity is generally higher in the $k-\varepsilon$ model due to its dependence on the turbulent dissipation rate (which can be relatively small in high turbulence regions), allowing for larger values. On the other hand, the $k-\omega$ model typically predicts a more tightly controlled eddy viscosity, especially near walls, due to its dependence on the specific dissipation rate, which tends to



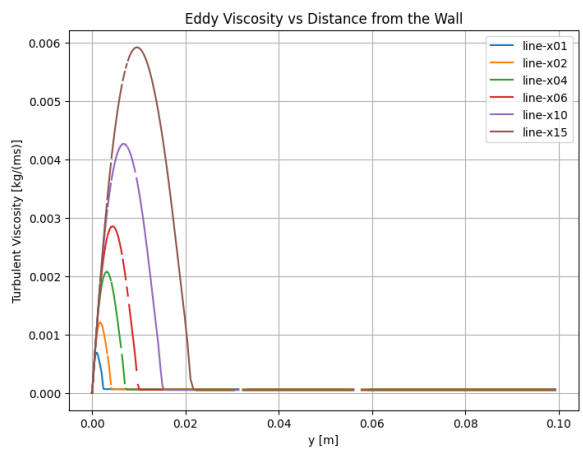
(a) Eddy Viscosity, ν_t, distribution for $k-\varepsilon$ turbulence model on coarse mesh.



(b) Eddy Viscosity, ν_t, distribution for $k-\varepsilon$ turbulence model on fine mesh.



(c) Eddy Viscosity, ν_t, distribution for $k-\omega$ turbulence model on coarse mesh.



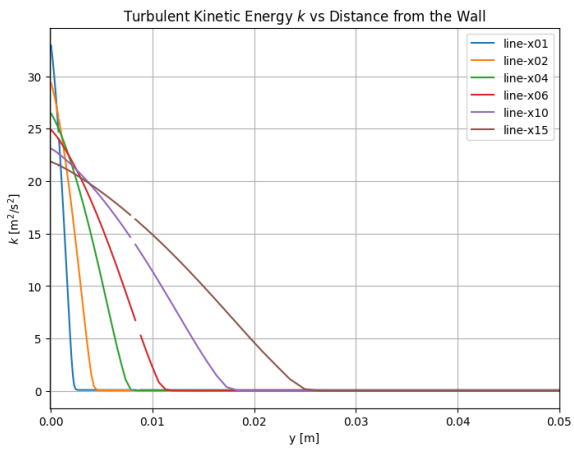
(d) Eddy Viscosity, ν_t, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 8: Eddy viscosity, μ_t, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

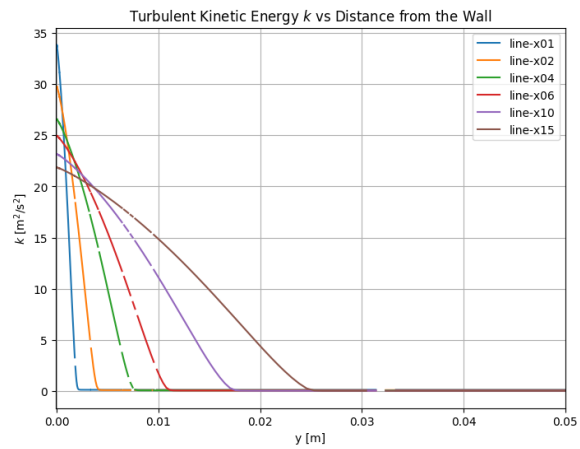
remain higher in such regions. Finer grid doesn't affect eddy viscosity calculation.

Turbulence kinetic energy distributions are presented on Fig. 9. In turbulent flows near solid boundaries, such as the walls of a pipe or the surface of a flat plate, the turbulence is strongly influenced by the interaction between the fluid and the wall. The boundary layer is a region of the flow where the velocity gradient is high due to the **no-slip condition** at the wall. Close to the wall, turbulent eddies are generated and break down at small scales due to friction. The turbulent kinetic energy is created in this region by the large vortical structures and dissipates into smaller scales as it moves away from the wall. Turbulent kinetic energy is highest close to the wall because of the intense shear stress and

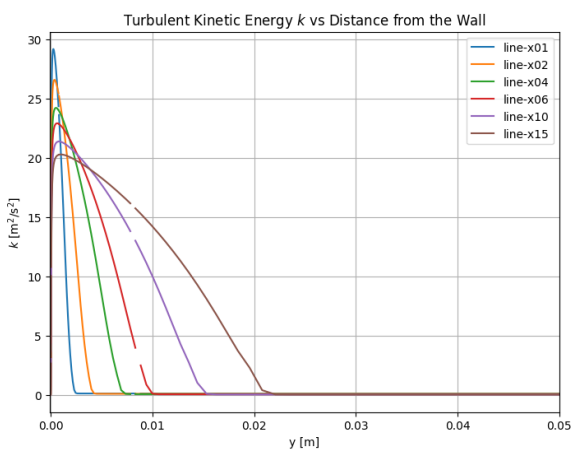
turbulence generation in that region. As you move away from the wall, the velocity gradients decrease, reducing the turbulence intensity. However, the energy from turbulent eddies takes time to dissipate, which leads to the slower decrease of turbulent kinetic energy as you move farther from the wall.



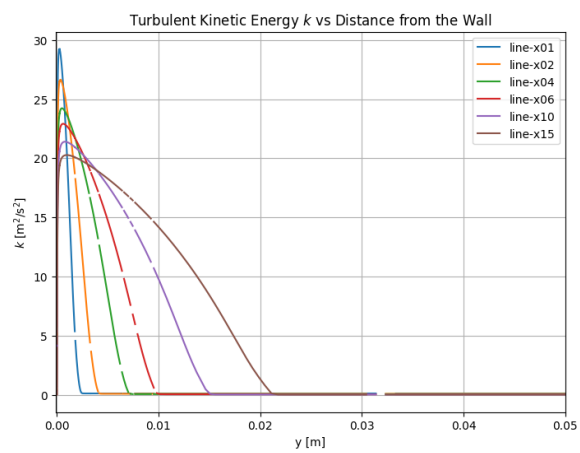
(a) Turbulence kinetic energy, k, distribution for $k-\varepsilon$ turbulence model on coarse mesh.



(b) Turbulence kinetic energy, k, distribution for $k-\varepsilon$ turbulence model on fine mesh.



(c) Turbulence kinetic energy, k, distribution for $k-\omega$ turbulence model on coarse mesh.



(d) Turbulence kinetic energy, k, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 9: Turbulence kinetic energy, k, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

Since the $k-\omega$ model is more sensitive to near-wall effects, k starts from zero at the wall, reflecting the absence of turbulence there due to the no-slip boundary condition. The turbulence kinetic energy increases as you move away from the wall, with the growth controlled by the ω term. In $k-\varepsilon$ model, the turbulent kinetic energy starts from a positive value at the wall because the model assumes that turbulence is present in the boundary layer (even close to the wall) and is generated by the shear

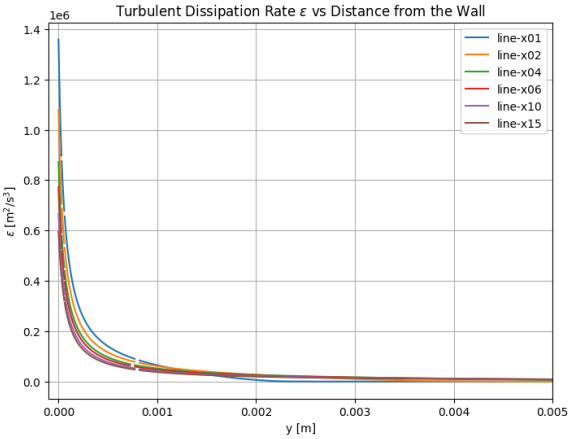
stress from the no-slip condition. Finer mesh produces little bit higher maximum turbulent kinetic energy, but the results for both meshes are consistent.

The turbulence dissipation rate distributions, ε, are presented on the Fig. 10. In both models, the dissipation rate ε peaks in the near-wall region where turbulence production is highest and then gradually decreases away from the wall. Farther from the wall, turbulence is more homogeneous, and large eddies dominate over smaller dissipative eddies. Since dissipation mainly occurs at the smallest turbulence scales, and these scales are reduced farther from the wall, ε decreases until it reaches zero. In $k - \varepsilon$ model, the wall function approximation assumes that turbulence is well-developed in the near-wall region, even though the mesh does not resolve the smallest eddies, so ε remains finite because k is nonzero at the wall. In $k - \omega$ model, the dissipation rate ε is zero at the wall because both k and ω are defined in such a way that the dissipation physically vanishes where turbulence is absent. The finer mesh didn't affect the calculation, and the results for different grids for both models are consistent.

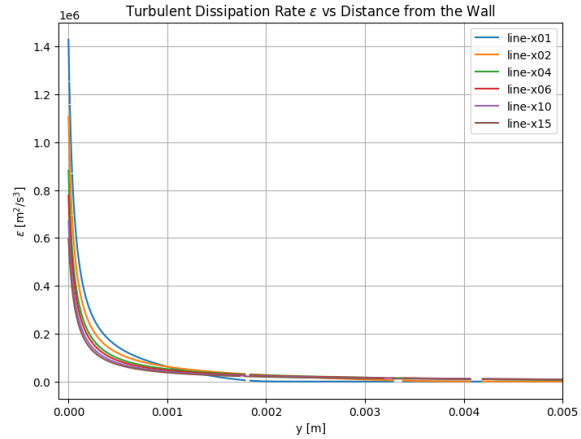
The distributions of specific dissipation rate are presented on Fig. 11. As it can be seen, the specific dissipation rate starts from a high value and rapidly decreases, and it is only present in $k - \omega$ model. Near the wall, ω needs to be very large because: k is very small (since turbulence production is limited by viscosity), ε is even smaller (since there is little turbulence dissipation in the viscous sublayer).

Since$\omega = \varepsilon/k$, if both ε and k are small but ε is relatively smaller, ω becomes very large. After moving slightly away from the wall, k increases faster than ε, so ω decreases very quickly. It can be observed that finer mesh produces higher initial value of the ω. The first cell height (y^+) in the mesh controls how well the boundary layer is resolved. For fine mesh the first grid point is very close to the wall, meaning the solver captures more of the steep gradients in turbulence quantities. Since ω is inversely proportional to the square of the near-wall distance, a smaller y directly leads to a higher initial value of ω. If the mesh is too coarse, the ω peak is underpredicted, leading to inaccurate turbulence modeling in the near-wall region. If the mesh is too fine, it may lead to numerical stiffness in the solver due to very high ω values.

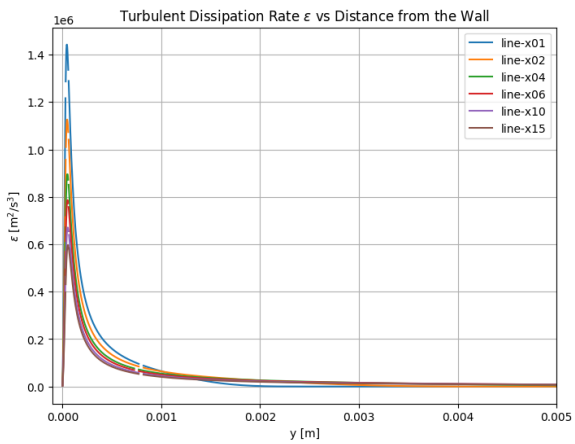
General remark: Through all the plots, the values plotted are taken at the certain x-position (x=0.1, 0.2, 0.4, 0.6, 1.0, 1.5). It will be evident that values associated with higher x value, will be higher. It can be interpreted by understanding that turbulence needs some distance to fully develop. At lower x values (closer to the inlet), turbulence is still growing, and ε and ω are lower. As x increases, the boundary layer thickens, shear increases, and turbulence production intensifies, leading to higher dissipation rates.



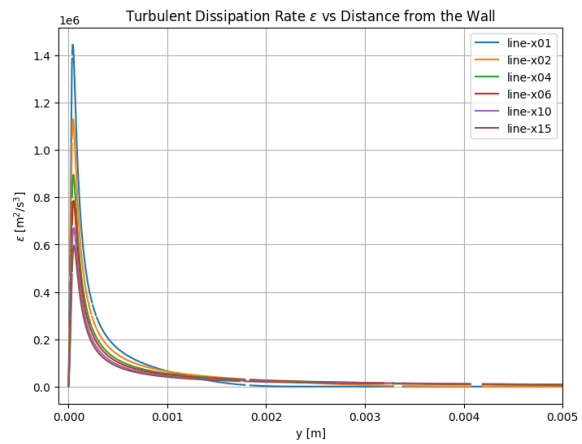
(a) Turbulence dissipation rate, ε, distribution for $k-\varepsilon$ turbulence model on coarse mesh.



(b) Turbulence dissipation rate, ε, distribution for $k-\varepsilon$ turbulence model on fine mesh.



(c) Turbulence dissipation rate, ε, distribution for $k-\omega$ turbulence model on coarse mesh.



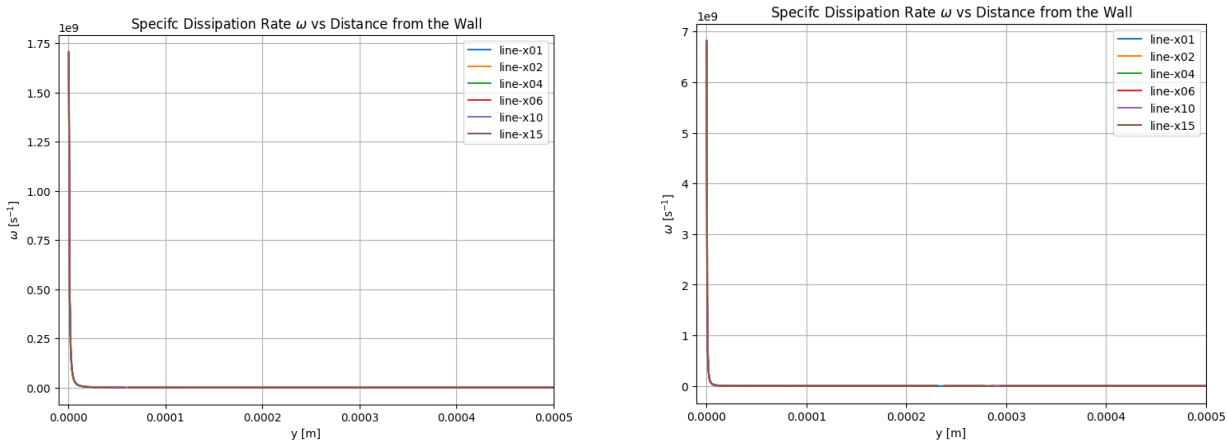
(d) Turbulence dissipation rate, ε, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 10: Turbulence dissipation rate, ε, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

4.2 Backward Step

The U_{ref} is the reference velocity at the center-channel near x/H=-4, used to nondimensionalize velocity and turbulent shear stress profiles. The skin friction coefficient and pressure coefficient data were also evaluated with respect to conditions near this location. Note, however, that when plotted the pressure coefficient data has been shifted uniformly so that C_p is 0 near the position x/H=0.4.

Simulation result of the pressure coefficient and friction coefficient are presented on Fig. 12a and 12b, respectively. Just after the step, the flow separates and a low-pressure recirculation zone



(a) Specific dissipation rate, ω, distribution for $k-\omega$ turbulence model on coarse mesh.

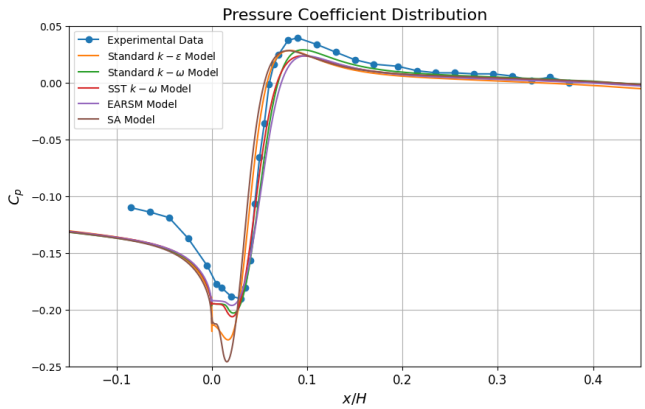
(b) Specific dissipation rate, ω, distribution for $k-\omega$ turbulence model on fine mesh.

Figure 11: Specific dissipation rate, ω, distributions for $k-\omega$ and $k-\varepsilon$ turbulence models for both grids.

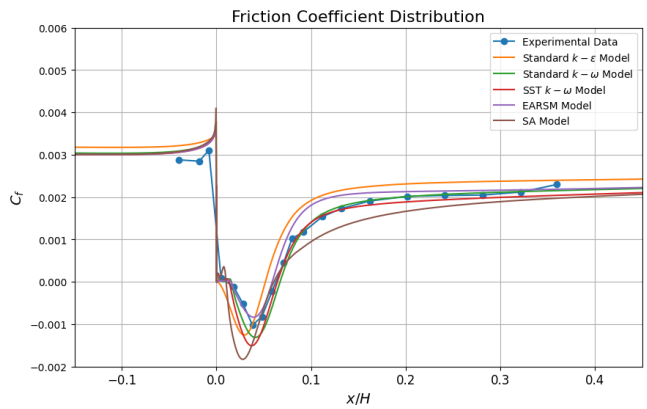
forms. C_p drops sharply due to local pressure drop. As the flow reattaches, pressure begins to recover downstream. From the Fig. 12a it can be visible that before the step and immediately after the step, theoretical predictions are underestimating the C_p coefficient. In recovering zone all five turbulence models represent physics devotedly comparing to the experimental results. The EARSM and standard $k-\omega$ models yield the closest match to C_p experimental data in the backward-facing step flow because it offers better near-wall turbulence modeling and accurately predicts the separation-induced pressure drop and recovery. Its robustness in adverse pressure gradients makes it especially suited for flows involving strong recirculation.

From the Fig. 12b can be seen that EARSM and standard $k-\omega$ in this case, as well, show the best alignment with C_f experimental data. Just after the step, flow separates - shear stress drops to nearly zero or becomes negative (reverse flow at the wall). Sudden increase in C_f indicates reattachment point. In the backward-facing step flow, analyzing C_f allows to assess the ability of turbulence models to resolve near-wall turbulence and predict the correct location of reattachment, which is essential for accurate flow modeling and design. The reattachment point marks the end of the recirculation zone where separated flow returns to the wall. Its accurate prediction is essential for capturing wall shear stress trends and evaluating turbulence model performance in separated flow regimes.

The reattachment points predicted by each simulation are presented in Table 1. Among the models, the SST $k-\omega$ model yields the reattachment point closest to the experimental measurements. The small variations in U_∞, p_∞, ρ_∞ arise due to differences in turbulence model behavior and how each affects the flow field during iterative convergence, even under identical boundary conditions.



(a) Pressure coefficient validation simulation results with different turbulence models utilized.



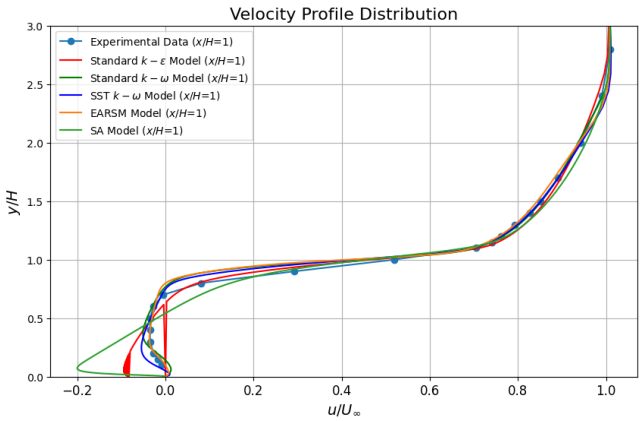
(b) Skin friction coefficient validation simulation results with different turbulence models utilized.

Model	U_∞ [m/s]	p_∞ [Pa]	ρ_∞ [kg/m^3]	x/H	C_f
Spalart-Allmaras (SA)	44.2429	99846.1	1.16325	0.060833	8.59E-06
Standard $k-\varepsilon$	44.1383	99851.1	1.16329	0.051667	2.53E-06
Standard $k-\omega$	44.2518	99846.1	1.16325	0.067292	4.06E-06
SST $k-\omega$	44.2431	99846.1	1.16325	0.064583	1.61E-05
EARSM	44.2432	99846.1	1.16325	0.0591667	1.94E-05
Experiment	/	/	/	0.0626 ± 0.0010	/

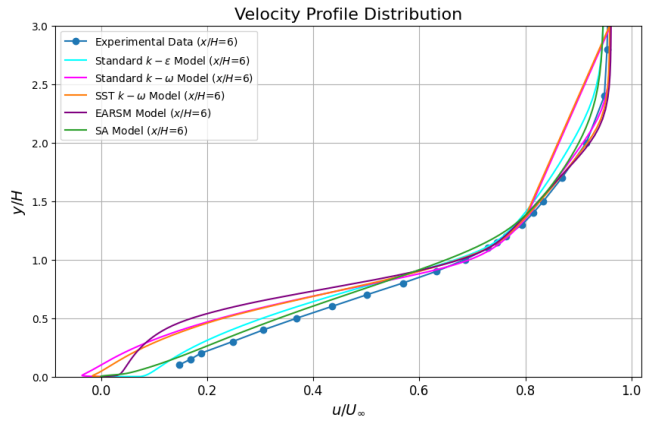
Table 1: Values of U_∞, p_∞, ρ_∞ and reattachment points in simulation cases.

Velocity profiles were observed at x/H=1 and x/H=6. Around x/H=1, depending on the Reynolds number and geometry, the flow begins to reattach. This is a critical zone where: turbulence production peaks, velocity gradients are high, models are pushed to capture both the separation and reattachment dynamics. Comparing velocity profiles and Reynolds stresses here helps assess how well each model captures the separated shear layer and reattachment. By x/H=6 point, the flow has likely reattached and begun to recover toward a new equilibrium boundary layer. This region is useful to check: whether the model can predict the redevelopment of the boundary layer post-reattachment, how well it estimates Reynolds stresses in the fully turbulent flow. It tells you how accurately the model captures the long-term effects of separation. Results are presented in Fig. 29 and 13b.

At the near-reattachment region ($x/H = 1$), the SST $k-\omega$ and standard $k-\omega$ models demonstrate better agreement with the experimental velocity profiles. This is because $k-\omega$ models are more effective at capturing adverse pressure gradients and flow separation, which are dominant in this region. In contrast, further downstream at $x/H = 6$, the standard $k-\varepsilon$ model provides a closer match to experimental data. This can be attributed to the model's strength in predicting fully developed turbulent flows, where the flow has reattached and redeveloped into an equilibrium boundary layer. Unusual



(a) Velocity distribution at $x/H = 1$ for different turbulence models compared with experimental data.



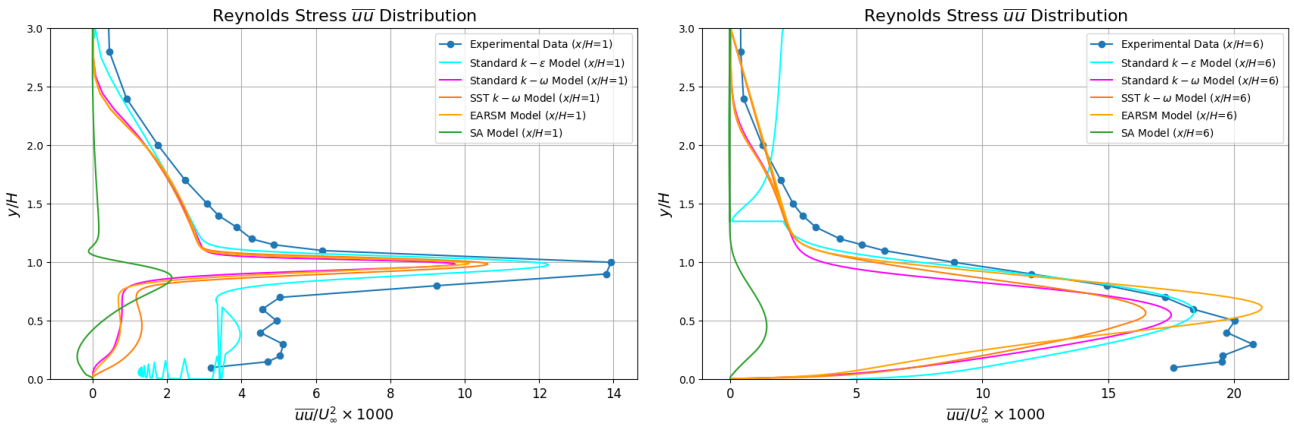
(b) Velocity distribution at $x/H = 6$ for different turbulence models compared with experimental data.

fluctuations on the graph are consequences of inadequate generation of file data before postprocessing.

Reynolds stresses represent the turbulent momentum transport in a flow. They are defined as the correlations of fluctuating velocity components, like $\overline{u'v'}$, and are essential in describing how turbulence affects the mean flow. One can simulate Reynolds stresses to evaluate how well turbulence models capture the physics of turbulent transport. In this study, the Mach number is low (Ma$\leq$0.129) - meaning density changes are negligible. The flow is not affected by temperature or pressure changes enough to change density. Most lab-scale backward-facing step experiments and simulations are in low-speed, constant-density conditions. Incompressible assumptions are made because the flow is low-speed, and this simplification doesn't hurt accuracy.

Results of Reynolds stress $\overline{uu}$ component distribution are presented on the Fig. 14a and 14b. The results of the Reynolds stress $\overline{uu}$ component distribution are presented in Fig. 14a and 14b. From Fig. 14a, it can be observed that the standard $k - \varepsilon$ model provides the best agreement with experimental data across the entire wall-normal range. However, in Fig. 14b, the same model fails to accurately predict the stress distribution for $y/H > 1$. Sometimes, $k - \varepsilon$ models tend to overpredict turbulence kinetic energy (TKE) in separated flows. This can accidentally align with experimental $\overline{uu}$ profiles in the near-field region, especially when the experimental peak is high. At $x/H = 6$, the flow is reattached and resembles a boundary layer recovering from separation - a tough region for the standard $k - \varepsilon$ model.

The SA model, being a one-equation eddy-viscosity model, typically underpredicts the peak of $\overline{u'u'}$ in this region. This is because the model is not specifically designed to capture the strong anisotropy and elevated turbulence levels associated with separated shear layers and reattachment zones. Its formulation, optimized for attached wall-bounded flows, lacks the complexity to accurately



(a) Reynolds stress $\overline{uu}$ component distribution at $x/H = 1$ for different turbulence models compared with experimental data.

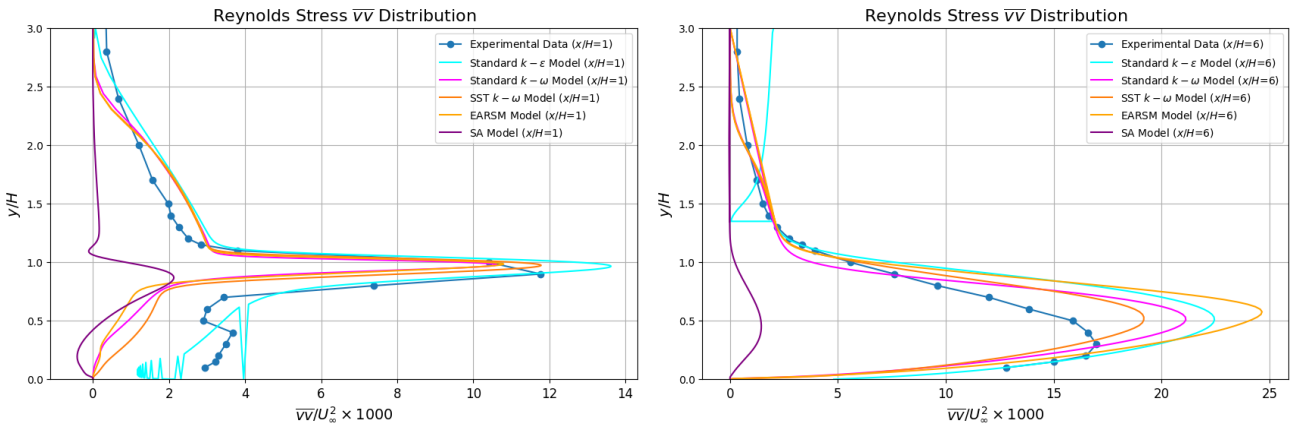
(b) Reynolds stress $\overline{uu}$ component distribution at $x/H = 6$ for different turbulence models compared with experimental data.

represent the rapid increase in turbulent fluctuations that occurs here. As a result, the SA model may show a smoother and lower peak compared to experimental or DNS data.

Further downstream, the flow has reattached and the boundary layer is redeveloping. In this recovery region, the SA model's prediction of $\overline{u'u'}$ often remains lower than experimental or DNS results, and the profile may appear too diffused. The model tends to underestimate the magnitude and spread of turbulent fluctuations because it lacks explicit treatment of turbulence transport and anisotropy, which are still relevant in the post-reattachment boundary layer. Studies show that even with calibration, the baseline SA model does not fully capture the persistence of elevated turbulence levels in the recovery zone.

Comparing both positions, the $\overline{uu}$ component exhibits a broader distribution at $x/H = 6$, indicating a more diffused turbulent field downstream. At $x/H = 1$, the stress peaks sharply around $y/H = 1$, which is typical near the shear layer formed just after separation. In both cases, the $\overline{uu}$ component remains positive, as it represents the normal stress due to streamwise velocity fluctuations, which by definition cannot be negative.

Results of $\overline{vv}$ component distribution of Reynolds stress is presented on Fig. 15a and 15b. Although the Reynolds stress distributions for $\overline{uu}$ and $\overline{vv}$ components show similar qualitative trends, they represent different physical processes. The streamwise component $\overline{uu}$ is typically larger in magnitude due to stronger velocity fluctuations along the main flow direction. The similarity in shape could be attributed to the isotropic assumptions inherent in the turbulence models, particularly in the standard $k - \varepsilon$ formulation, which may not fully capture the anisotropic nature of turbulence near separation and reattachment zones. At $x/H = 6$, where the flow has reattached and transitioned into



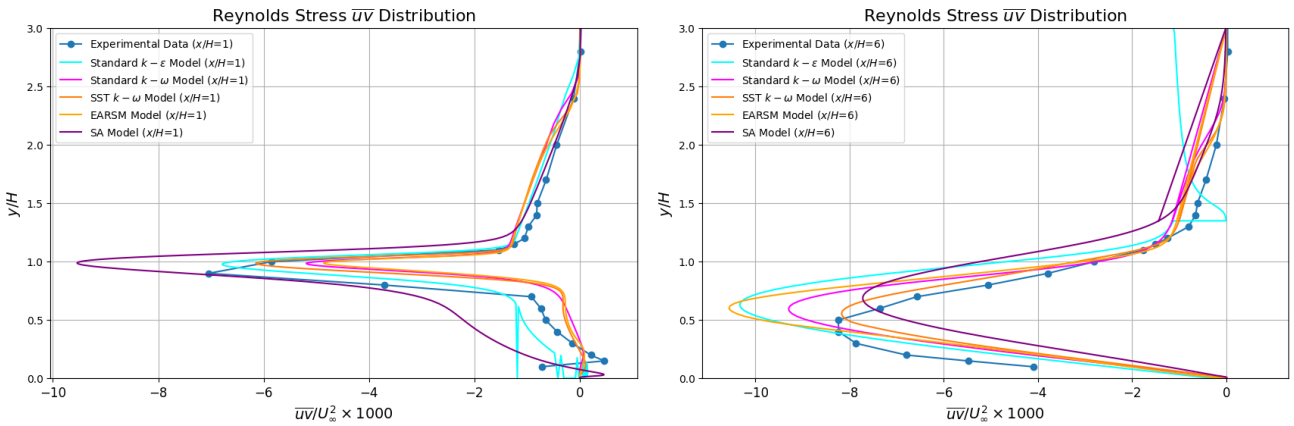
(a) Reynolds stress $\overline{vv}$ component distribution at $x/H = 1$ for different turbulence models compared with experimental data.

(b) Reynolds stress $\overline{vv}$ component distribution at $x/H = 6$ for different turbulence models compared with experimental data.

a developing turbulent boundary layer, all turbulence models tend to overpredict the peak of the $\overline{uu}$ Reynolds stress. Moreover, the predicted peak is often smeared compared to the experimental data, indicating a broader and less accurate distribution of turbulent energy in the streamwise direction.

This behavior is typical of RANS-based two-equation models, which rely on the assumption of eddy viscosity and isotropy. These models struggle to accurately capture the complex anisotropic turbulence structures and history effects present in reattaching flows. As a result, they tend to diffuse the turbulence too much, leading to overpredicted and smeared Reynolds stress peaks, particularly downstream of the reattachment point. The strong underestimation of $\overline{v'v'}$ by the Spalart-Allmaras model is due to its isotropic eddy-viscosity assumption and lack of explicit modeling for turbulence anisotropy, making it less suitable for predicting detailed Reynolds stress distributions in separated and reattaching flows. The significant overprediction of $\overline{v'v'}$ by the EARSM model at $x/H = 6$ is likely due to its strong sensitivity to turbulence anisotropy and possible limitations in the calibration of redistribution and near-wall damping mechanisms for this flow scenario. While EARSM generally improves the representation of complex turbulent stresses, it can, in some cases, overshoot the wall-normal component in the recovery region behind a backward-facing step.

A comparison of the Reynolds normal stress component $\overline{vv}$ at positions $x/H = 1$ and $x/H = 6$ reveals that the peak magnitude is significantly smaller at $x/H = 1$ than at $x/H = 6$. This behavior can be attributed to the undeveloped nature of wall-normal turbulence shortly after reattachment. Near the reattachment point, the turbulence is highly anisotropic, with streamwise fluctuations dominating. As the flow progresses downstream to $x/H = 6$, the turbulence becomes more isotropic and fully developed, resulting in stronger wall-normal velocity fluctuations and a corresponding increase in the $\overline{vv}$ stress component.



(a) Reynolds stress $\overline{uv}$ component distribution at $x/H = 1$ for different turbulence models compared with experimental data.

(b) Reynolds stress $\overline{uv}$ component distribution at $x/H = 6$ for different turbulence models compared with experimental data.

Results of $\overline{uv}$ component of Reynolds stress is presented in Fig. 16a and 16b. In both cases, the predicted distributions are consistently negative, which aligns with physical expectations, as the negative $\overline{uv}$ represents turbulent momentum transfer from high-speed to low-speed regions in the wall-normal direction. However, at $x/H = 6$, all turbulence models overpredict the magnitude of $\overline{uv}$, especially in the region away from the wall. This overestimation is likely due to the turbulence models generating excess turbulent viscosity, leading to an exaggerated shear stress. This issue is common in RANS-based models, which assume isotropic turbulence and often fail to accurately capture the complex stress recovery in reattached boundary layers. The SA model overestimates the peak of the $\overline{u'v'}$ distribution compared to experimental data. This overprediction occurs because the SA model, as a one-equation eddy-viscosity model, tends to produce an overly strong eddy viscosity in regions of high shear and turbulence production - such as the separated shear layer just before and during reattachment. The EARSM model underestimates the Reynolds shear stress near reattachment, likely due to limitations in capturing the rapid production and redistribution of turbulence in highly non-equilibrium regions. Further downstream, it tends to overestimate $\overline{u'v'}$, reflecting a tendency to sustain higher turbulence levels than observed, possibly due to the model's strong sensitivity to anisotropy and a less accurate representation of turbulence decay in the post-reattachment boundary layer.

The Reynolds shear stress component $\overline{uv}$, representing turbulent momentum transport, shows distinct behavior at different streamwise locations. Experimental data indicates that the peak value of $\overline{uv}$ at $x/H = 6$ increases slightly, comparing to peak at $x/H = 1$, reflecting enhanced turbulent mixing as the flow redevelops after reattachment. Among the turbulence models evaluated, the SST $k - \omega$ model demonstrates the best agreement with experimental data at $x/H = 6$, accurately capturing both the peak magnitude and the shape of the shear stress distribution. This performance can be attributed

to the SST model's ability to blend the near-wall accuracy of the $k-\omega$ formulation with the free-shear behavior of the $k-\varepsilon$ model, making it particularly well-suited for flows involving separation and reattachment.

4.3 Subsonic Jet Plate

The Mach number distribution at the outlet was examined to confirm the subsonic nature of the jet, Fig. 17. The maximum Mach number obtained was approximately 0.525, indicating that the flow remains within the subsonic regime. This validates the use of a pressure-based solver, which is appropriate for low-speed, incompressible or mildly compressible flows.

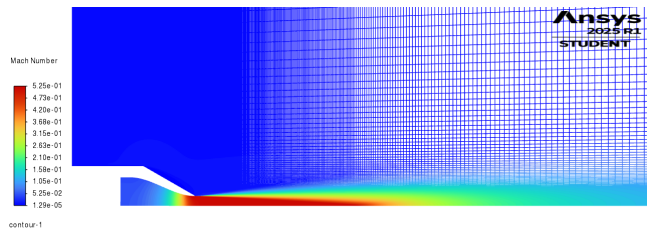
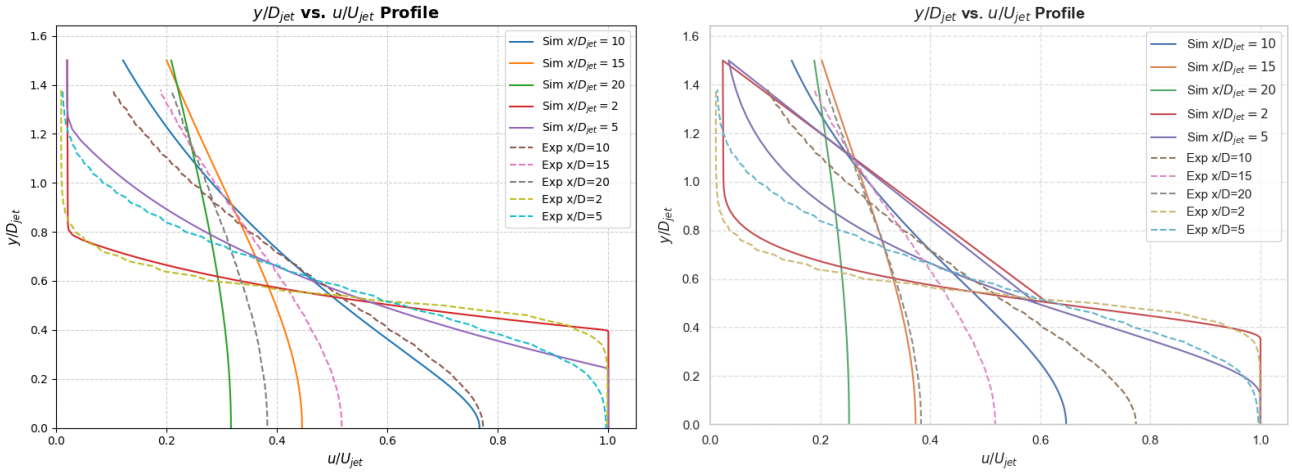


Figure 17: Mach number distribution at the outlet for 5% jet inlet turbulence intensity.

Simulation result of u/U_{jet} axial velocity distribution is plotted on Fig. 18. The axial velocity profiles at downstream positions $x/D_{jet} = 2$, 5, 10, 15, and 20 show a progressive broadening of the jet with increasing x/D_{jet}, indicative of jet spreading and momentum diffusion into the surrounding fluid. At $x/D_{jet} = 2$, the velocity profile is still relatively sharp, resembling a top-hat shape characteristic of a potential core region. As the flow evolves downstream, the profiles transition toward a self-similar Gaussian-like shape, with reduced peak velocity and increased radial spread.

Interestingly, across all tested turbulence intensities (1%, 5%, and 15%), the velocity profiles were nearly identical at all streamwise locations. This indicates that the SST $k-\omega$ model, within the context of this simulation, is relatively insensitive to variations in inlet turbulence intensity. The turbulent mixing and shear layer development appear to be primarily governed by the flow dynamics rather than the specified inlet turbulence level. In the near field ($x/D_{jet} = 2$), the CFD results show good agreement with experimental data, both in the shape and magnitude of the velocity profile. This suggests the SST $k-\omega$ model effectively captures the initial development of the jet shear layer.

However, at larger downstream positions ($x/D_{jet} \geq 5$), noticeable discrepancies begin to emerge. The simulation under-predicts the axial velocity, particularly near the centerline region ($y/D_{jet} \approx 0$),



(a) Sim = SST $k - \omega$ model results.

(b) Sim = Spalart-Allmaras model results.

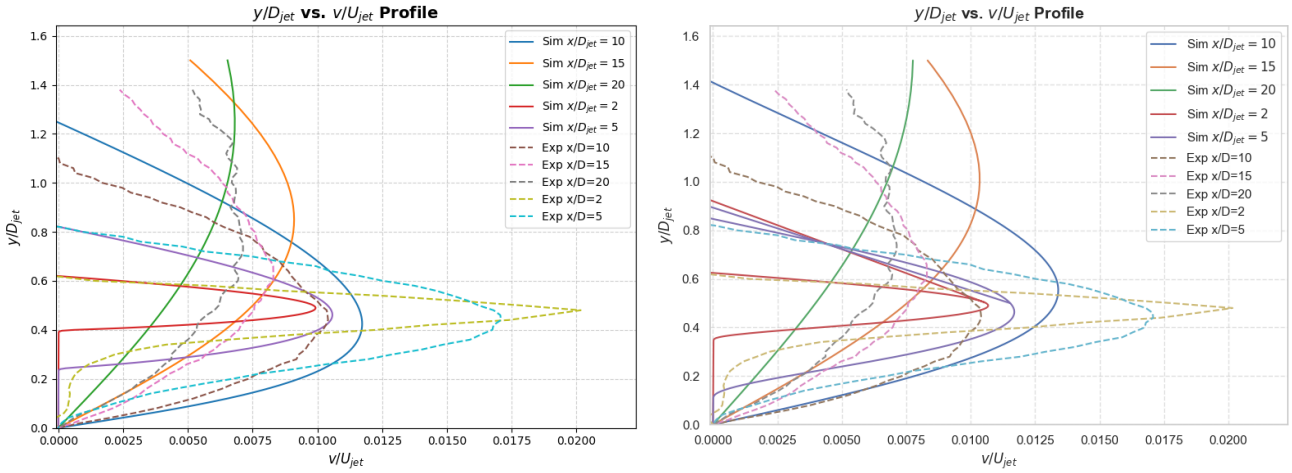
Figure 18: Axial velocity profile distribution for 5% jet inlet turbulence intensity.

where the experimental data show a higher velocity peak. This underestimation becomes more pronounced as x/D_{jet} increases, indicating that the model fails to sustain the core velocity as effectively as observed in the experiment.

The insensitivity of the SST model to inlet turbulence intensity-evidenced by nearly identical results for 1%, 5%, and 15% inlet turbulence - may contribute to this deviation. While the model self-generates turbulence and adequately captures the initial mixing, it may not correctly model the long-range momentum diffusion or turbulence decay processes that influence far-field jet behavior. These results suggest that while the SST $k - \omega$ model is suitable for near-field jet predictions, its accuracy degrades further downstream, particularly in replicating the centerline velocity distribution. This highlights a potential limitation in the model's predictive capability for fully developed jet flow unless more refined turbulence modeling or transition specification is introduced.

Simulation result of v/U_{jet} radial velocity distribution is presented on Fig. 19. The normalized radial velocity profiles provide insight into the entrainment behavior of the jet as it develops downstream. In contrast to axial velocity, which is dominant along the centerline, radial velocity peaks away from the centerline and reflects the outward movement of fluid driven by shear layer expansion and turbulent mixing.

Across all streamwise locations, the radial velocity profiles from the simulation exhibit the characteristic behavior of a turbulent jet: low values near the centerline ($y/D_{jet} \approx 0$), with a gradual rise to a peak at intermediate (y/D_{jet}), followed by a decay toward the outer region. This peak corresponds to the location of the shear layer, where turbulent mixing is most intense.



(a) Sim = SST $k - \omega$ model results.

(b) Sim = Spalart-Allmaras model results.

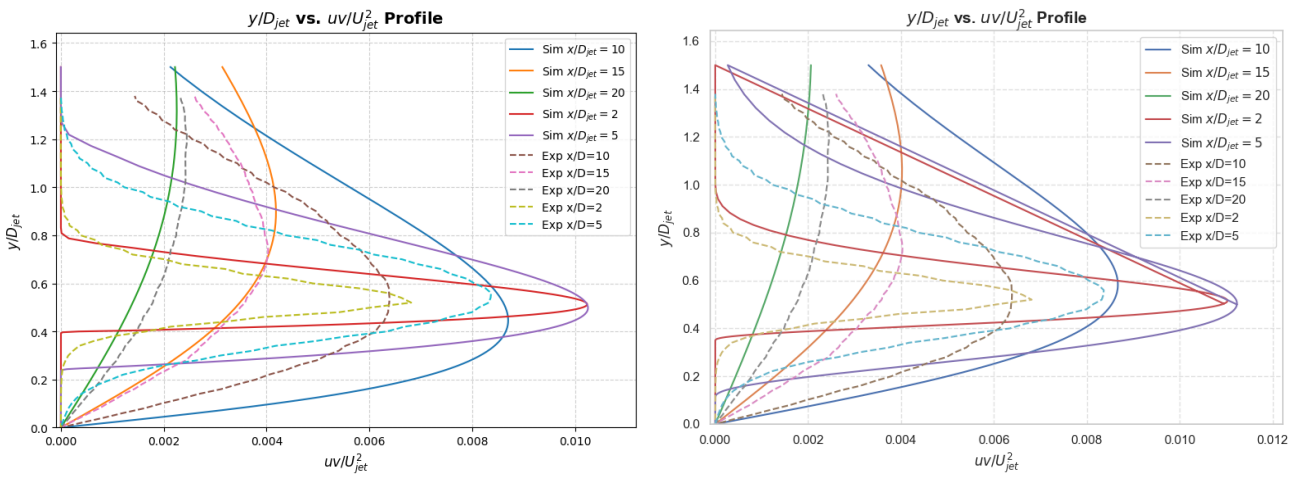
Figure 19: Radial velocity profile distribution for 5% jet inlet turbulence intensity.

However, when compared to the experimental data, the simulated radial velocities are generally underpredicted at most locations, particularly in the far field. This underestimation is consistent with the axial velocity discrepancies, suggesting a general trend where the model does not fully capture the extent of turbulent mixing and momentum transfer in the outer regions of the jet. The shape of the profile is similar to the experimental curves, but the magnitude of v/U_{jet} is lower. An exception occurs at x/D_{jet}, where the simulation overpredicts the radial velocity, especially in the region near the peak ($y/D_{jet} \approx 1$). This overprediction could be due to a localized amplification of shear layer activity in the model, possibly related to how the SST model balances the k and ω transport near that axial location. It's also possible that numerical diffusion or mesh resolution in that region influenced the velocity gradients and contributed to the spike.

Moreover, similar to the axial velocity results, the radial velocity profiles show negligible sensitivity to the inlet turbulence intensity. The fact that 1%, 5%, and 15% turbulence intensity cases produce nearly identical results implies that the SST model generates internal turbulence rapidly, overshadowing the effects of prescribed inlet conditions. Overall, while the SST model qualitatively reproduces the structure of the radial velocity profiles, it appears to underestimate the strength of lateral entrainment, particularly further downstream. This could be a consequence of limited turbulent diffusion or energy transfer in the model, again pointing to possible improvements through higher-fidelity turbulence models or better inlet turbulence specification.

Simulation of Reynolds shear stress distribution $u'v'/U_{jet}^2$ is presented on Fig. 20. The Reynolds shear stress profiles provide a direct measure of turbulent momentum transport across the jet, driven

primarily by velocity fluctuations in the axial and radial directions. In typical turbulent jet flows, $u'v'/U_{jet}$ exhibits a peak near the shear layer and approaches zero near the centerline and at large radial distances, consistent with reduced turbulence activity in these regions.



(a) Sim = SST $k-\omega$ model results.

(b) Sim = Spalart-Allmaras model results.

Figure 20: Reynolds shear stress distribution for 5% jet inlet turbulence intensity.

The simulation results show qualitatively correct behavior: at all streamwise locations, the Reynolds shear stress peaks at intermediate y/D_{jet}, near the edge of the potential core, and gradually diminishes toward the jet axis and far field. The location of the peak shifts outward with increasing x/D_{jet}, reflecting the radial growth of the turbulent shear layer. The shape of the distribution is very similar to that of the radial velocity, highlighting the physical link between turbulent mixing and lateral momentum transport.

However, the quantitative agreement with experimental data varies significantly across axial locations:

- At x/D_{jet}=2, 5, and 10, the simulation overpredicts the Reynolds shear stress throughout most of the shear layer. This suggests that the SST $k-\omega$ model generates stronger-than-observed turbulent momentum transport in the near and mid-field regions, likely due to its formulation being aggressive in capturing early shear layer development.
- At x/D_{jet}=15, the simulation underpredicts the shear stress for $y/D_{jet} < 0.7$, especially near the centerline. Beyond that point, the simulation begins to overpredict again in the outer shear layer. This reversal may point to a shift in the model's estimation of eddy viscosity or turbulence production away from the jet core.

- At x/D_{jet}=20, the model consistently underpredicts the Reynolds shear stress across the entire radial range. This underestimation aligns with similar trends seen in axial and radial velocity at far downstream locations, indicating that the SST model struggles to sustain turbulence intensity and mixing as the jet becomes more diffused.

Again, across all turbulence intensity cases (1%, 5%, 15%), the profiles remain effectively unchanged, reinforcing the earlier observation that inlet turbulence intensity has little influence on the predicted turbulence fields in the SST model for this configuration. The turbulence is likely generated primarily by the flow gradients at the nozzle exit and in the shear layer, overpowering the effect of the specified inlet conditions. In summary, while the SST $k-\omega$ model reproduces the general shape and location of the Reynolds shear stress profiles, it systematically underpredicts the magnitude of turbulent momentum transport when compared to experimental data, with limited sensitivity to inlet turbulence levels.

When comparing the Spalart-Allmaras (SA) and SST $k-\omega$ turbulence models for a subsonic 2D jet plate against theoretical or experimental data, you will notice that both models can capture the general features of the jet, but there are important differences in their predictions and underlying modeling approaches. Both SA and SST $k-\omega$ models reproduce the overall velocity and Reynolds stress profiles of the jet, including the spreading of the jet and the evolution of mean and fluctuating velocities downstream. This is evident from plots above, where both models follow the experimental trends for u/U_{jet}, v/U_{jet}, and $\overline{uv}/U_{jet}^2$ across y/D_{jet} at different x/D_{jet} locations. For many practical engineering flows, both models provide reasonable agreement with experimental data, especially in the near field and for mean velocity predictions.

Spalart-Allmaras (SA) is a one-equation model designed primarily for attached, wall-bounded flows. It solves a single transport equation for a modified eddy viscosity and assumes turbulence is isotropic. SST $k-\omega$ is a two-equation model that blends the k-omega model near the wall (for better accuracy in boundary layers) with the k-epsilon model in the free stream. It is more sensitive to adverse pressure gradients and separation, and generally handles complex flows better.

SA Model tends to be less accurate in free shear flows (like jets) because it does not directly solve for turbulent kinetic energy (k) or dissipation, and its eddy viscosity formulation can be too diffusive or insufficiently responsive to the anisotropy of turbulence in jets. This can lead to under- or over-prediction of Reynolds stresses ($\overline{uv}$), especially further downstream where turbulence decays and anisotropy matters more. SST $k-\omega$ by solving for both k and ω, it can better capture the evolution and decay of turbulence, leading to improved predictions of Reynolds stresses and jet spreading rates, especially as the flow develops downstream

Both models capture the reduction of centerline velocity and the spreading of the jet, but SST k-omega typically predicts the jet width and velocity decay more accurately in line with experiments, especially at larger x/D_{jet}. As seen on the Fig. 18, the SST k-omega model often matches the experi-

mental $\overline{uv}$ profile more closely, especially in the outer shear layer and at larger downstream distances, while SA can overpredict or underpredict depending on the region and flow development.

At smaller x/D_{jet} (closer to the jet exit), both models are quite similar and align well with experimental data. At larger x/D_{jet}, differences become more pronounced. The SA model may overpredict turbulent shear stress ($\overline{uv}$) and cross-stream velocity (v) in the outer region, while SST k-omega maintains better agreement with experiments, especially for velocity decay and jet width. These differences are due to the SST k-omega model's better handling of turbulence transport and decay, and its ability to model anisotropy and turbulent diffusion more accurately in free shear flows.

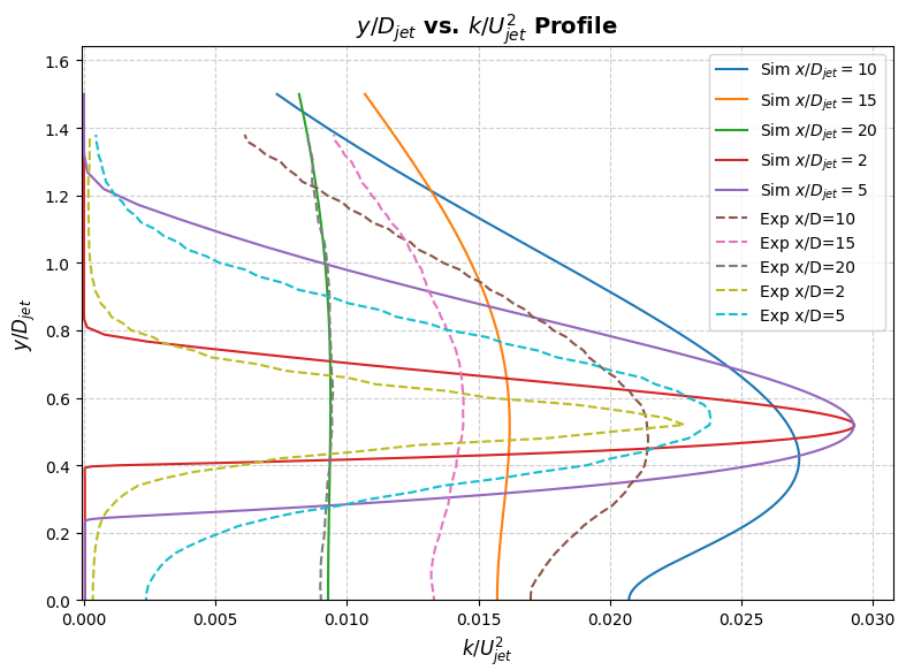


Figure 21: Turbulence kinetic energy distribution for 5% jet inlet turbulence intensity. Sim = SST $k-\omega$ model results.

Simulation result of turbulence kinetic energy k is given on Fig. 21. The distribution of turbulent kinetic energy provides direct insight into the turbulence intensity and how energy is partitioned among fluctuating velocity components across the jet. Spalart-Allmaras model cannot reproduce this distribution since it doesn't resolve the turbulence kinetic energy k.

In all examined downstream locations, the simulation captures the expected qualitative features: low TKE along the centerline (=0), with a peak occurring within the shear layer, followed by a decline toward the jet's outer edge. This reflects the physical reality that turbulence is primarily generated in

regions with high velocity gradients, especially at the boundary between the jet core and the ambient fluid.

However, the quantitative behavior of the simulation relative to experimental data varies with axial position:

- For x/D_{jet} = 2, 5, 10, 15 simulation overpredicts the turbulent kinetic energy across much of the radial profile. This overprediction aligns with earlier observations from the Reynolds shear stress distribution, where stronger turbulent mixing was also simulated in the near and mid-field regions.
- For x/D_{jet} = 20: The simulation matches the experimental turbulent kinetic energy throughout the radial profile.

The simulation results also show that inlet turbulence intensity has negligible effect on the predicted TKE profiles. All three cases (1%, 5%, and 15%) collapse onto nearly identical distributions, reinforcing that the model-generated turbulence dominates over the specified inlet conditions. In summary, while the SST $k - \omega$ model captures the general structure of the TKE distribution, it tends to overpredict turbulence levels in the near and mid-field and underpredict them in the far field, following a similar pattern observed in the Reynolds shear stress. These trends highlight the model's tendency to generate excessive turbulence early on, while potentially over-dissipating it farther downstream.

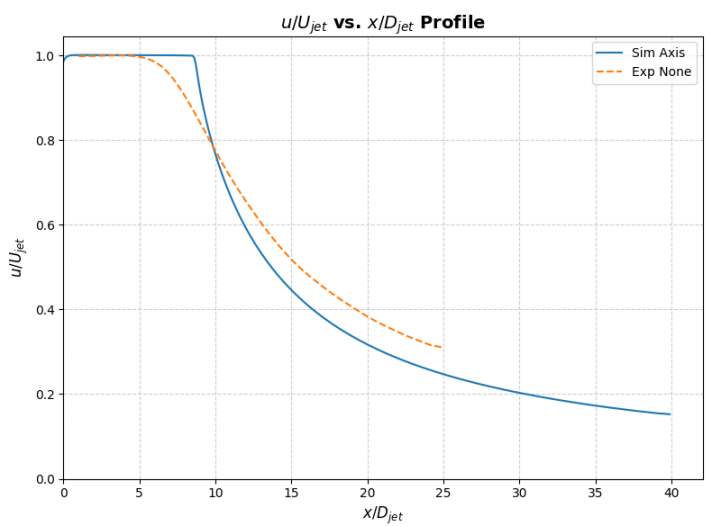


Figure 22: Axial velocity profile over axis for 5% jet inlet turbulence intensity.

The axial velocity profile u/U_{jet} decreases as the jet moves downstream, following typical jet development behavior. In the near-field (x/D_{jet}= 2,5), the simulation provides a reasonable match

to experimental data, with high velocity near the centerline. In the far-field (x/D_{jet}=15, 20), discrepancies arise as the simulation tends to underpredict the velocity, possibly due to limitations in the turbulence model's ability to capture the full extent of jet expansion and dissipation. Overall, the distribution reflects the jet's natural tendency to lose momentum and spread radially with distance.

4.4 Turbulent Mixing Layer

Figure 23 shows the velocity magnitude distribution across the 2D mixing layer domain. Two distinct freestream regions are clearly visible:

- The upper stream with a higher velocity (around 42 m/s),
- The lower stream with a lower velocity (around 17 m/s).

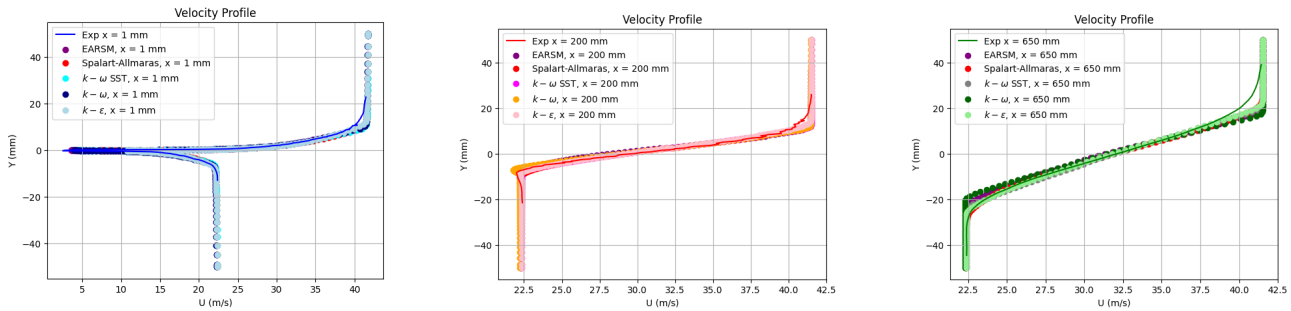
At the interface between these two streams, a mixing layer develops, characterized by a gradual transition of velocity values between the two streams. The thickness of this shear region increases as the flow progresses downstream, illustrating the growth of the mixing layer due to turbulent diffusion. The overall distribution is smooth and symmetric, consistent with expectations for a mixing layer under nearly incompressible flow conditions (low Mach number). No visible numerical instabilities or nonphysical oscillations are observed, indicating that the simulation setup and boundary conditions were properly defined.

Figure 23: Velocity magnitude distribution over domain.

4.4.1 Velocity profile at different streamwise locations

In the simulation of the mixing layer, velocity profiles were extracted at three streamwise locations: $x = 1$ mm, 200 mm, and 650 mm. The profiles show significant evolution as the flow develops downstream as depicted on Fig. 24.

At x = 1 mm, Fig. 24a, very close to the inlet, the two streams have not yet mixed significantly. The velocity profile exhibits a sharp gradient, closely resembling a step change between the two freestream velocities, indicating a very thin shear layer with minimal turbulent mixing. For x = 200 mm,



(a) Velocity profile at $x = 1$ mm streamwise location.

(b) Velocity profile at $x = 200$ mm streamwise location.

(c) Velocity profile at $x = 650$ mm streamwise location.

Figure 24: Velocity profiles for three different streamwise locations.

Fig. 24b, the shear layer has grown thicker due to the effects of turbulent mixing. The velocity transition between the two streams becomes smoother, and the flow starts to exhibit typical turbulent mixing behavior, with a broader region of intermediate velocities.

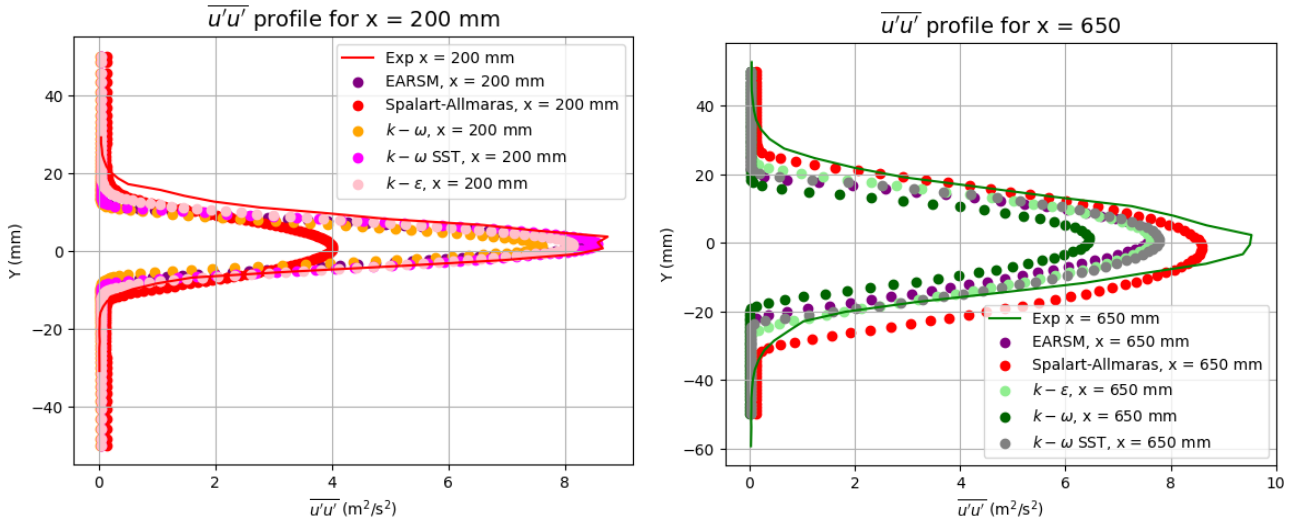
Considering x = 650 mm, the shear layer has thickened even further, and the velocity profile becomes more self-similar, meaning its shape, when properly normalized, does not change significantly with further downstream distance. The centerline region shows a much smoother gradient, indicating a high degree of mixing between the two streams. The differences between the profiles at these three locations are primarily due to the growth of the mixing layer caused by turbulent momentum transfer, which spreads the velocity differences over an increasingly wider region as the flow progresses downstream.

The results show that all five turbulence models predict similar velocity profiles, and they all capture the general behavior and development of the mixing layer well. Small differences can be observed in the sharpness and thickness of the velocity gradient, but overall, each model provides acceptable agreement with theoretical expectations and experimental measurements. This suggests that, for the prediction of mean velocity in a turbulent mixing layer, the choice between $k - \varepsilon$, $k - \omega$, SST $k - \omega$, Spalart-Allmaras and EARSM models is not critical, and all models are capable of delivering reasonable accuracy.

4.4.2 Reynolds stress components at different streamwise locations

Reynolds stress profiles were extracted at $x = 200$ mm and $x = 650$ mm to analyze the development of turbulent fluctuations within the mixing layer.

At x = 200 mm, Fig. 25a, the $\overline{u'u'}$ distribution shows a sharp peak centered within the mixing layer, indicating intense turbulence production where the velocity gradient is largest. The peak is rel-



(a) Reynolds stress $\overline{u'u'}$ component distribution at $x =$ 200 mm streamwise location.

(b) Reynolds stress $\overline{u'u'}$ component distribution at $x =$ 650 mm streamwise location.

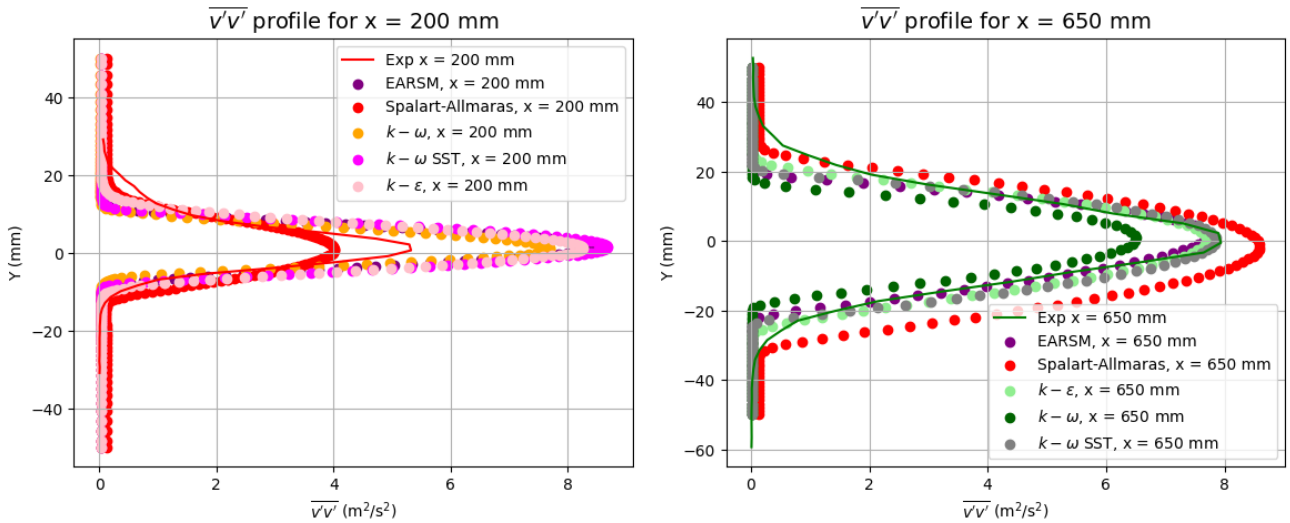
Figure 25: Reynolds stress $\overline{u'u'}$ component distribution at two different streamwise locations.

atively high and the distribution remains narrow, reflecting the strong shear-driven turbulence near the upstream region. The predicted distributions from all three turbulence models show good agreement with the experimental data. Among the models, the SST $k-\omega$ and EARSM models provide the closest match to the experimental peak value, while the standard $k-\varepsilon$ and $k-\omega$ models slightly underpredict the peak. However, the discrepancies are small, and overall all models perform reasonably well at this location. Spalart-Allmaras (SA) eddy visocsity totally underestimate the peak value. The SA model's lack of shear-specific calibration and isotropic assumption explain its underestimation of peak $\overline{u'u'}$.

At x = 650 mm, Fig. 25b, the mixing layer has broadened, and the $\overline{u'u'}$ distribution becomes wider and flatter. The peak value increased compared to x = 200 mm. This reflects the continued growth of turbulence as the mixing layer evolves downstream, with larger turbulent structures developing and increased turbulent kinetic energy. These observations are consistent with the expected behavior of a turbulent shear layer, where turbulence production initially dominates and is followed by gradual diffusion and dissipation of turbulent kinetic energy further downstream. All three turbulence models underpredict the Reynolds stress levels compared to the experimental measurements. The EARSM, SST $k-\omega$ and $k-\varepsilon$ models show very similar behavior, with their profiles closely overlapping each other. The standard $k-\omega$ model underpredicts the peak even more significantly, resulting in the largest discrepancy among the three. This suggests that while the models are able to reasonably capture the early development of turbulence, they tend to underestimate the strength of turbulence fluctuations farther downstream, possibly due to limitations in modeling the growth of large-scale turbulent structures in a mixing layer. Turbulence transitions to a decay-dominated regime,

where dissipation and diffusion become significant. SA's overly diffusive eddy viscosity formulation artificially sustains turbulence levels, causing overprediction of $\overline{u'u'}$.

The turbulence models perform well at x = 200 mm because the flow is still in the early development phase, characterized by strong shear-driven turbulence production, relatively isotropic turbulence structures, and a thinner mixing layer. These conditions align closely with the assumptions behind eddy-viscosity-based RANS models, leading to good agreement with experimental Reynolds stress measurements.



(a) Reynolds stress $\overline{v'v'}$ component distribution at $x =$ 200 mm streamwise location.

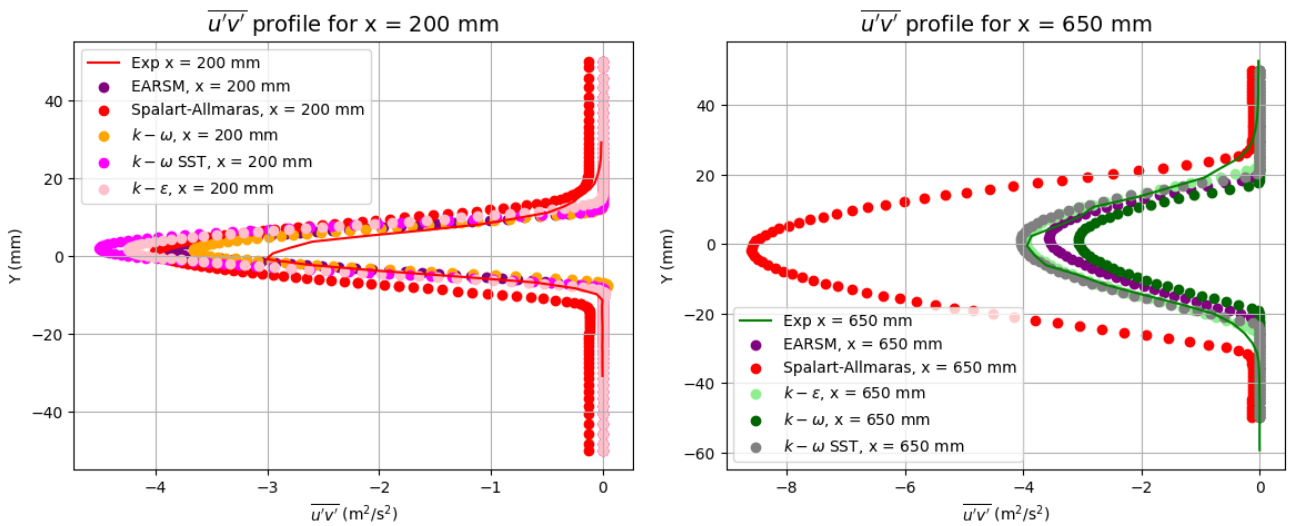
(b) Reynolds stress $\overline{v'v'}$ component distribution at $x =$ 650 mm streamwise location.

Figure 26: Reynolds stress $\overline{v'v'}$ component distribution at two different streamwise locations.

The Reynolds stress component $\overline{v'v'}$ represents the fluctuating velocity correlation in the vertical direction. At $x = 200$ mm, Fig. 26a, the results show a distinct peak in the Reynolds stress profile near the shear layer's center, which is typical for turbulent flows where high velocity gradients exist. The peak is relatively high and the distribution remains narrow, reflecting the strong shear-driven turbulence near the upstream region. But the peak is smaller comparing to the $\overline{u'u'}$ distribution of Reynolds stress, indicating weaker turbulence development in that direction. Even though, in this case, all three two-equation and EARSM turbulence models don't show huge discrepancies between themselves, they tend to over-predict the Reynolds stress. But Spalart-Allmaras again underestimates the peak value due to its formulation of homogenous tubrulent viscosity.

At $x = 650$ mm, Fig. 26b,the mixing layer has broadened, and the $\overline{v'v'}$ distribution becomes wider and flatter. The peak value increased compared to x = 200 mm. This reflects the continued growth

of turbulence as the mixing layer evolves downstream, with larger turbulent structures developing and increased turbulent kinetic energy. But compared to the $\overline{u'u'}$ distribution the peak is smaller and it can be attributed to smaller gradient, i.e. weaker turbulence development in y direction. In this case, EARSM, $k-\varepsilon$ and $k-\omega$ SST models accurately match the experimental values, while the $k-\omega$ underestimates the distribution and Spalart-Allmaras overestimates it. This can be atributed to that the $k-\omega$ model uses the turbulence frequency ω and is designed to resolve the near-wall region better by capturing the small turbulent structures. However, in a growing mixing layer, the larger turbulent structures that develop in the far-field are not as well captured by the $k-\omega$ model. The SST model, which is a hybrid of both the $k-\omega$ and $k-\varepsilon$ models, improves this by transitioning to the $k-\varepsilon$ model farther from the wall, where the larger turbulent eddies are more significant, thus providing a better match with the experimental data.



(a) Reynolds stress $\overline{u'v'}$ component distribution at $x =$ 200 mm streamwise location.

(b) Reynolds stress $\overline{u'v'}$ component distribution at $x =$ 650 mm streamwise location.

Figure 27: Reynolds stress $\overline{u'v'}$ component distribution at two different streamwise locations.

The distribution of $\overline{u'v'}$ represents the Reynolds stress associated with the covariance between the velocity components in the $x-$ and y-directions, often referred to as the shear stress in turbulence. In the context of a mixing layer, the behavior of this component is influenced by the shear in the flow and the interactions between the turbulent structures in the x and y directions.

At $x = 200$ mm, the $\overline{u'v'}$ component exhibits a sharp gradient across the mixing layer, with negative values indicating the momentum transfer between the turbulent structures in the horizontal and vertical directions. This negative correlation is typical for turbulent shear layers, where fluctuations in the u-direction (horizontal velocity) are anticorrelated with those in the v-direction (vertical velocity). The negative values suggest that turbulent eddies are transferring momentum from one stream to the

other, leading to shear stress across the layer. The profile shows a well-defined region of negative shear stress near the center of the mixing layer, where the turbulence is most intense, and gradually becomes less negative towards the outer regions of the layer. All five turbulence models overpredict the values of Reynolds shear stress because they tend to overestimate the turbulence intensity in the initial stages of the mixing layer. This is often due to the models' assumption of isotropic turbulence or the inability to fully capture the complexity of the shear layer's development, particularly in the early stage where turbulence is still growing. The models may overestimate the magnitude of velocity fluctuations and shear stress due to insufficient resolution of the turbulent structures near the initial shear zone. As a result, the predicted $\overline{u'v'}$ is higher than what is observed experimentally.

At $x = 650$ mm, the $\overline{u'v'}$ component becomes more spread out, reflecting the further evolution of the mixing layer and the growth of turbulent structures. The distribution at this downstream location is wider and less steep, as the turbulence continues to develop and the mixing layer grows. Despite the broader profile, the negative values of $\overline{u'v'}$ persist, indicating that the turbulent mixing between the two streams continues to be dominated by shear stress interactions. The magnitude of the negative $\overline{u'v'}$ is slightly increased compared to $x = 200$mm. The profile at this location indicates that the turbulence is spreading across a larger area, leading to a reduction in the intensity of the shear stress in the $\overline{u'v'}$-direction. SST $k - \omega$, $k - \epsilon$ and EARSM models match the experimental results accurately because they better capture the evolution of the turbulence in the downstream mixing layer, including larger eddies and reduced wall effects. However, the $k - \omega$ underestimates the shear stress because it is more sensitive to near-wall regions and struggles to accurately predict the turbulence in the wider, more developed flow, leading to a weaker prediction of the Reynolds shear stress. The Spalart-Allmaras model overestimates the $\overline{u'v'}$ Reynolds shear stress because it does not properly model the decay of turbulence in free shear flows, leading to excessive eddy viscosity and sustained turbulence levels far downstream. This is a known limitation of the standard SA model in mixing layers and similar free shear flow environments.

4.5 Rectangular Duct

The DNS result, Fig. 28, shows a complex secondary flow structure with small counter-rotating vortices in each corner of the duct cross-section, which is characteristic of low-Re turbulent duct flow.

These corner vortices are a result of imbalance in Reynolds stresses and pressure gradients near the walls.

Among the RANS models tested, the BSL EARSM model was the only one that qualitatively captured secondary flows, Fig. 29a. However, it produced only two large vortices sweeping flow towards the upper right corner, failing to reproduce the detailed four-corner vortex structure seen in the DNS. This is likely due to the model's ability to resolve some anisotropy in the Reynolds stress tensor, but still being limited in its representation of the finer turbulence structures.

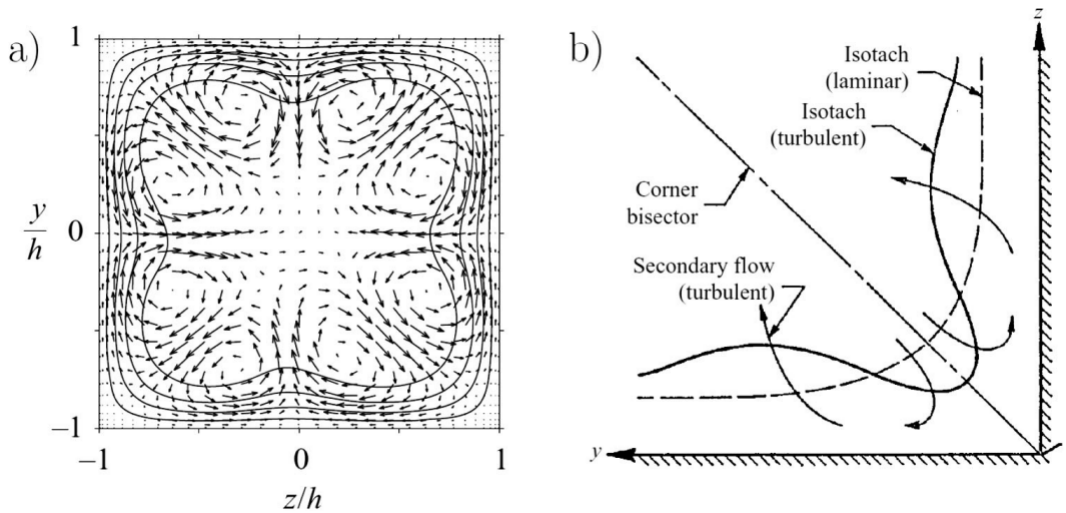
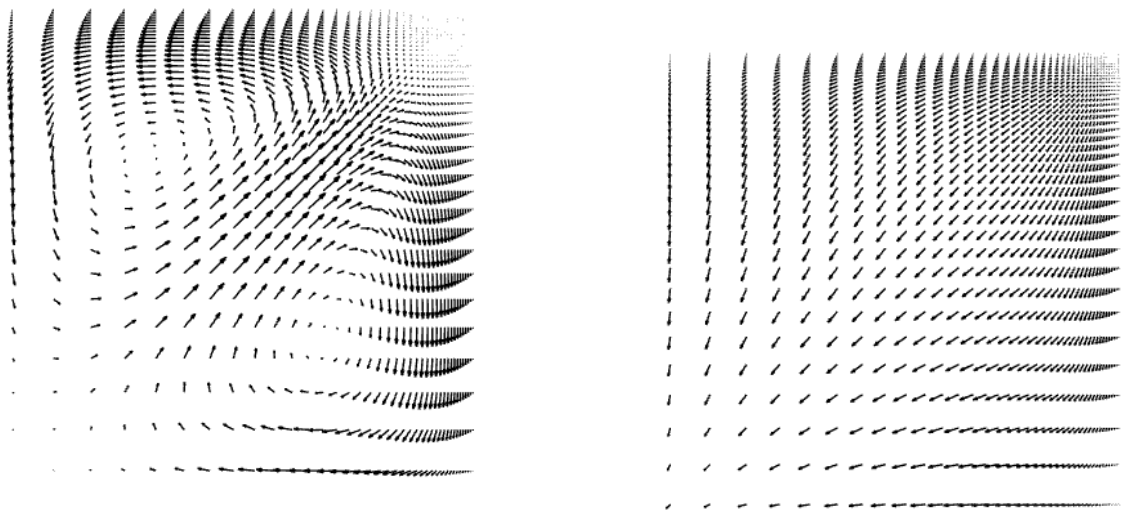


Figure 28: (a) Contour lines of the primary mean flow and vectors of the secondary mean flow for a DNS at a Reynolds number based on bulk velocity Re = 1205. (b) Sketch of the mean flow in the corner region of a straight duct. Taken from [**?**].

The other models - Spalart-Allmaras, SST $k - \omega$, and the Zero Equation (Mixing Length) model - all produced similar, oversimplified flow patterns, Fig. 29b, 30a and 30b. In these cases, velocity vectors showed a dominant single-directional motion towards the lower left corner, with little or no indication of secondary flow development in other corners. The fact that the vector scale had to be increased to visualize the motion suggests that these models underpredicted the transverse velocity components, likely due to their isotropic eddy-viscosity assumptions and lack of Reynolds stress anisotropy modeling.

Only the BSL EARSM model captured secondary flow structures, albeit in a simplified form with two dominant vortices, while the simpler eddy-viscosity models failed to resolve the corner vortices observed in the DNS and showed flow biased toward a single direction. The failure of the Spalart-Allmaras, SST, and mixing-length models to predict secondary flow arises from their reliance on isotropic eddy-viscosity assumptions. These models cannot capture turbulence-induced anisotropy or the cross-stream Reynolds stress gradients that drive secondary motion in ducts.

The failure of the Spalart-Allmaras, SST, and mixing-length models to predict secondary flow arises from their reliance on isotropic eddy-viscosity assumptions. These models cannot capture turbulence-induced anisotropy or the cross-stream Reynolds stress gradients that drive secondary motion in ducts. The need to manually increase the velocity vector scale in the SA, SST, and zero-equation cases further illustrates the models' inability to generate substantial secondary velocity components, confirming that the predicted flows were predominantly unidirectional.



(a) Secondary flow vectors at the duct outlet predicted using the BSL EARSM model. The model captures two dominant counter-rotating vortices, indicative of partial resolution of secondary flows due to its anisotropy-aware formulation.

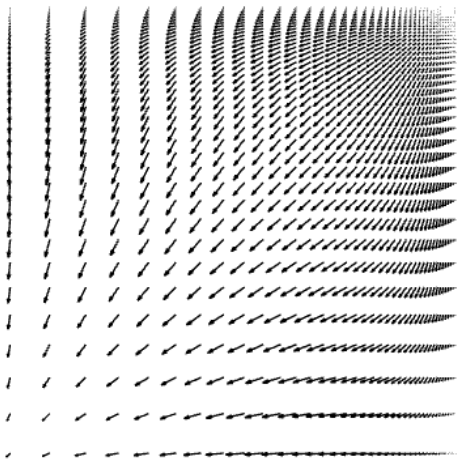
(b) Secondary flow vectors at the duct outlet predicted using the SST $k - \omega$ model. The flow pattern is largely unidirectional, with weak indications of secondary motion and poor representation of corner vortices.

Figure 29: Flow vectors at the duct outlet for BSL EARSM and SST $k - \omega$ models.

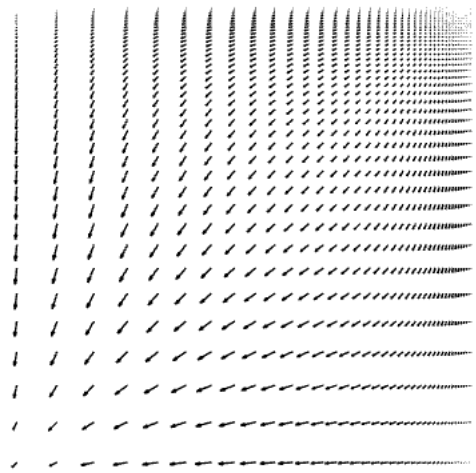
For engineering applications where secondary flows influence heat transfer or particle deposition (e.g., in cooling channels or clean rooms), relying on simple turbulence models may lead to inaccurate design predictions due to their inability to resolve these critical flow features.

Another trend observed in the outlet velocity vector plots is that the maximum velocity magnitude decreased progressively with more advanced turbulence models. For instance, the mixing-length model and Spalart-Allmaras showed higher peak velocities near the wall or lower left corner regions, while the SST model predicted slightly lower peak values. The BSL EARSM model yielded the lowest maximum vector magnitudes among all RANS simulations.

This behavior can be attributed to the increased physical realism and diffusivity of the more advanced models. Simpler models often overpredict turbulence mixing, leading to exaggerated secondary motion or spurious peak velocities. In contrast, models like SST and especially EARSM account for more accurate near-wall damping and turbulence anisotropy, leading to a more diffused and realistic velocity field with smoother gradients and lower peaks.



(a) Secondary flow vectors at the duct outlet using the Spalart-Allmaras model. The flow shows predominantly one-directional behavior, with minimal transverse motion and no evidence of secondary vortices.



(b) Secondary flow vectors at the duct outlet using the Mixing Length (Zero-Equation) model. The solution shows a simplified flow structure lacking any secondary flow features or corner vortices.

Figure 30: Flow vectors at the duct outlet for one-equation Spalart-Allmaras and zero-equation mixing-length models.

The reduced vector magnitudes in more advanced models may also reflect a suppression of unphysical flow features that simpler models are unable to filter out due to their limitations in handling complex 3D effects and low-Re behavior.

4.5.1 Convergence times

The convergence times for the four turbulence models vary significantly, as shown in Table 2. The Zero-Equation and Spalart-Allmaras models converged the fastest, requiring only 57 and 58 iterations respectively, with total simulation times of approximately 3 minutes. These models are relatively simple and involve solving fewer equations, with Spalart-Allmaras being a one-equation model and the Zero-Equation model using algebraic closure without solving any additional PDEs.

The SST $k - \omega$ model required 107 iterations, taking longer in terms of both iteration count and average time per iteration due to its two-equation structure and additional complexity near walls. The BSL EARSM model took the longest to converge (131 iterations, $\sim$6.5 minutes), despite a slightly faster per-iteration time than SST. This can be attributed to the model's increased mathematical complexity - particularly the algebraic Reynolds stress model - which adds stiffness to the solution and

Model	Average time per iteration [s]	Total time [s]	Number of iterations
BSL EARSM	2.956	387.201	131
SST	3.501	374.622	107
Spalart-Allmaras	2.873	166.611	58
Zero-Equation	3.218	185.417	57

Table 2: Comparison of convergence performance for the four turbulence models. The table shows the average time per iteration, total simulation time, and number of iterations required for each model to converge.

makes convergence more challenging. These results confirm the expected trade-off between model fidelity and computational cost: simpler models converge faster but lack physical detail, while more advanced models like BSL EARSM provide better accuracy at the expense of longer simulation times and more iterations.

5 Conclusion

This project comprehensively evaluated the performance of various turbulence models-Mixing Length, Spalart-Allmaras (SA), One-Equation, Standard and SST k-ω, Standard k-ε, and EARSM-across diverse benchmark test cases representing both wall-bounded and free-shear turbulent flows.

For the **flat plate** case, the k-ω model demonstrated superior near-wall accuracy, especially on coarse grids, due to its favorable behavior in the viscous sublayer. The Mixing Length model significantly overpredicted wall friction coefficients, while the One-Equation model offered a balanced trade-off between accuracy and complexity. Grid sensitivity analysis revealed that finer meshes do not always guarantee better results, particularly when wall treatment is mismatched with mesh resolution.

In the **backward-facing step** simulation, models capable of resolving adverse pressure gradients and separation-particularly SST k-ω and EARSM-showed the best agreement with experimental reattachment locations, wall shear stress, and pressure recovery. Velocity and Reynolds stress profiles validated the limitations of simpler models like SA in resolving separated shear layers and turbulence anisotropy.

The **subsonic jet** analysis highlighted the SST k-ω model's strong performance in the near field but decreasing accuracy downstream. The model's insensitivity to inlet turbulence intensity suggests dominant turbulence generation at the nozzle exit. Discrepancies in centerline velocity and underprediction of shear stress and turbulence kinetic energy in the far field indicate a limitation of RANS models in sustained free-shear turbulent development.

In the **turbulent mixing layer**, all tested models captured the general development of the shear layer. SST k-ω, EARSM, and k-ε models more closely matched experimental Reynolds stress distributions, especially at early streamwise locations. The Spalart-Allmaras model consistently underperformed due to its simplistic, isotropic eddy-viscosity assumption, while the EARSM model showed better capability to capture anisotropic turbulence features but occasionally overpredicted turbulent quantities.

The **rectangular duct** simulation underscored the necessity of anisotropy-aware modeling. Only the EARSM model partially captured secondary flows, producing two main vortices, while all other models predicted unidirectional flows. This limitation arises from the inability of conventional eddy-viscosity models to resolve turbulence-induced secondary motions caused by cross-stream stress gradients.

In terms of **computational performance**, simpler models like SA and Mixing Length converged faster, whereas the EARSM and SST models required more iterations and longer wall-clock times due to increased physical fidelity and equation complexity.

Final Remarks

This study demonstrates that no single turbulence model universally outperforms others across all flow regimes. While simpler models offer rapid solutions suitable for preliminary engineering applications, more advanced models such as SST k-ω and EARSM provide improved predictive accuracy

in complex flow features like separation, anisotropy, and secondary motion-albeit with increased computational cost. Future work could involve hybrid RANS-LES methods or transition-sensitive models to further improve fidelity in flows with strong non-equilibrium turbulence behavior.

References

[1] Inc. ANSYS. *Spalart-Allmaras Model*, 2025.

[2] Inc. ANSYS. *SST k-omega Model*, 2025.

[3] Inc. ANSYS. *Standard k-epsilon Model*, 2025.

[4] Inc. ANSYS. *Standard k-omega Model*, 2025.

[5] Inc. ANSYS. *WJ-BSL-EARSM k-omega Model*, 2025.

[6] Peter Davidson. *Turbulence: an introduction for scientists and engineers*. Oxford university press, 2015.

[7] Garem J. H. Delville J., Bellin S. and Bonnet J. P. 2dml: 2d mixing layer validation case. `https://turbmodels.larc.nasa.gov/delvilleshear_val.html`, 1989.

[8] David M. Driver and Harold L. Seegmiller. 2dbfs: 2d backward facing step. `https://turbmodels.larc.nasa.gov/jetsubsonic_val.html`, 1985.

[9] David M. Driver and Harold L. Seegmiller. 2dbfs: 2d backward facing step. `https://turbmodels.larc.nasa.gov/backstep_val.html`, 1985.

[10] DC Wilcox. Turbulence modeling for cfd. *DCW industries, La Canada*, 1998.